

Research Methods

Collect & Analyze Finance Data from LSEG with Python (2025/2026)

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Pr. Alexandre Garel



- Professeur de finance durable, Audencia
- Chercheur (climat, biodiversité, AI, politique de vote, finance comportementale)
- Expert dans des comités scientifiques (fonds biodiversité)
- Expert qualifié (FIR – Forum de l'Investissement Responsable)
- Membre du HKUST Global Sustainable Institute
- Missions de conseils (KPMG...)

Pr. Alexandre Garel

Google Scholar



Audencia

Alexandre Garel

MAIN ABOUT PUBLICATIONS WORKING PAPERS PRESS DATA TEACHING

CV

FULL PROFESSOR · FINANCE, AUDENCIA BUSINESS SCHOOL

CLIMATE FINANCE BIODIVERSITY FINANCE ESG BEHAVIORAL FINANCE CORPORATE FINANCE AI

Google Scholar LinkedIn SSRN Email

NAVIGATE RESEARCH & WORK



About

PUBLICATIONS

Firm-Level Nature Dependence [Review of Finance](#)

Mutual Fund Voting on Climate [Mgmt. Science](#)

Employee Diversity & Performance [Fin. Management](#)

Shareholder Engagements in Banks [Econ. Letters](#)

Climate Risk & Mutual Fund Flows [J. Banking & Fin.](#)

Publications

WORKING PAPERS

Climate Disclosure & Director Elections [R&R](#)

Women-Led Mutual Funds Voting [R&R](#)

Climate Risk Engagements

Mandatory Carbon Disclosure

From Vision to Prediction (AI Pipeline)

Working Papers

Summary

1. Refinitiv/LSEG
2. Collecting data with Python
3. Graph & charts with Python
4. Descriptive statistics with Python
5. T-test with Python
6. Regression analysis with Python
7. Event study with Python
8. Portfolio analysis with Python
9. Cursor

1. Refinitiv/LSEG

Install Refinitiv **Workplace**:

<https://cdn.refinitiv.com/Apps/ProductDownloadPage/1.1.13/>



Welcome to **Refinitiv Workplace**

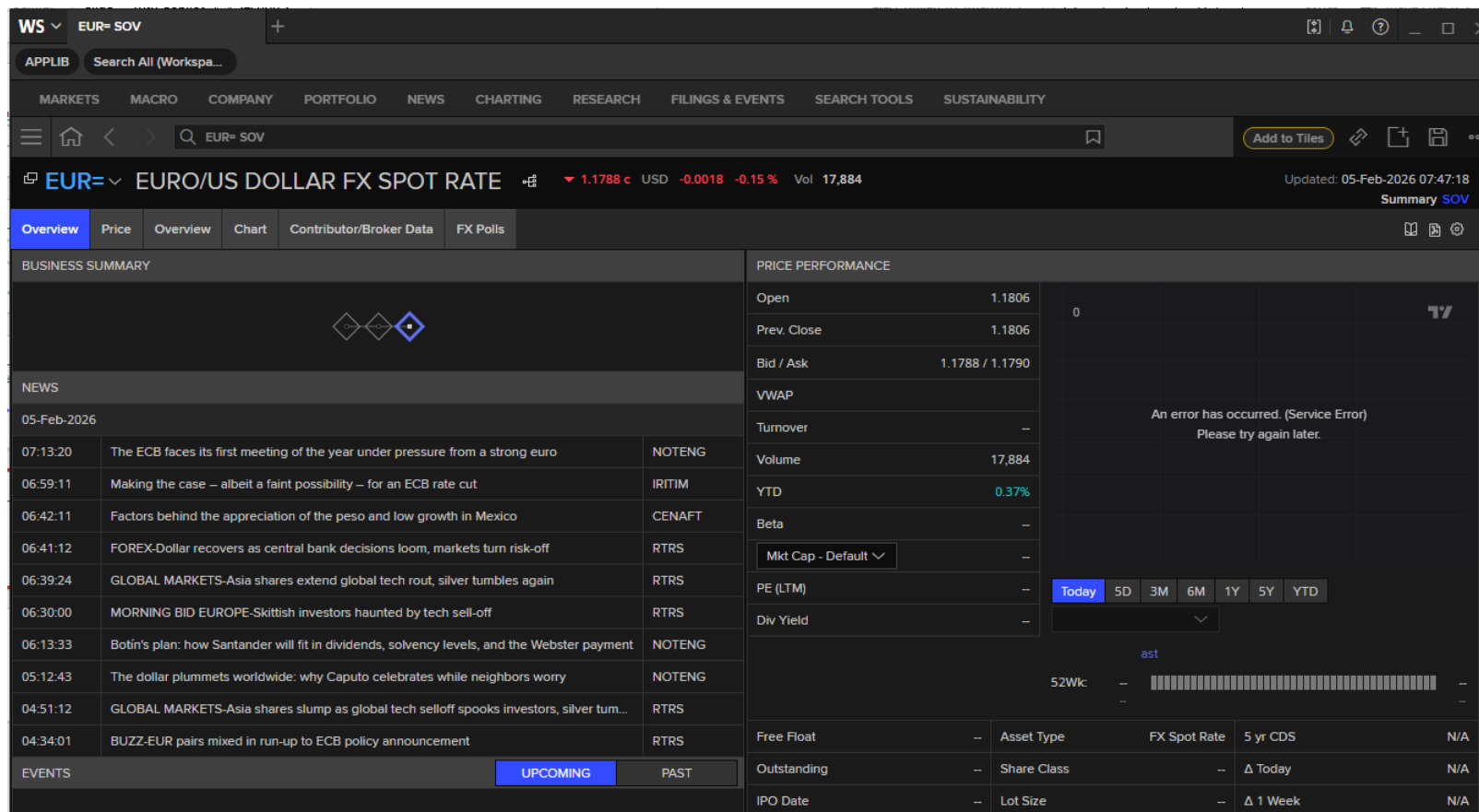
Redesigned for you. With you.

Help us reshape Refinitiv Workplace for anything the future may bring.

Download for Windows

Refinitiv Workplace is also available on [macOS](#) for you to download and install.

1. Refinitiv/LSEG



Knowledge Hub sent you your login and password, shared among several students

Cannot have more than one student using a shared account simultaneously

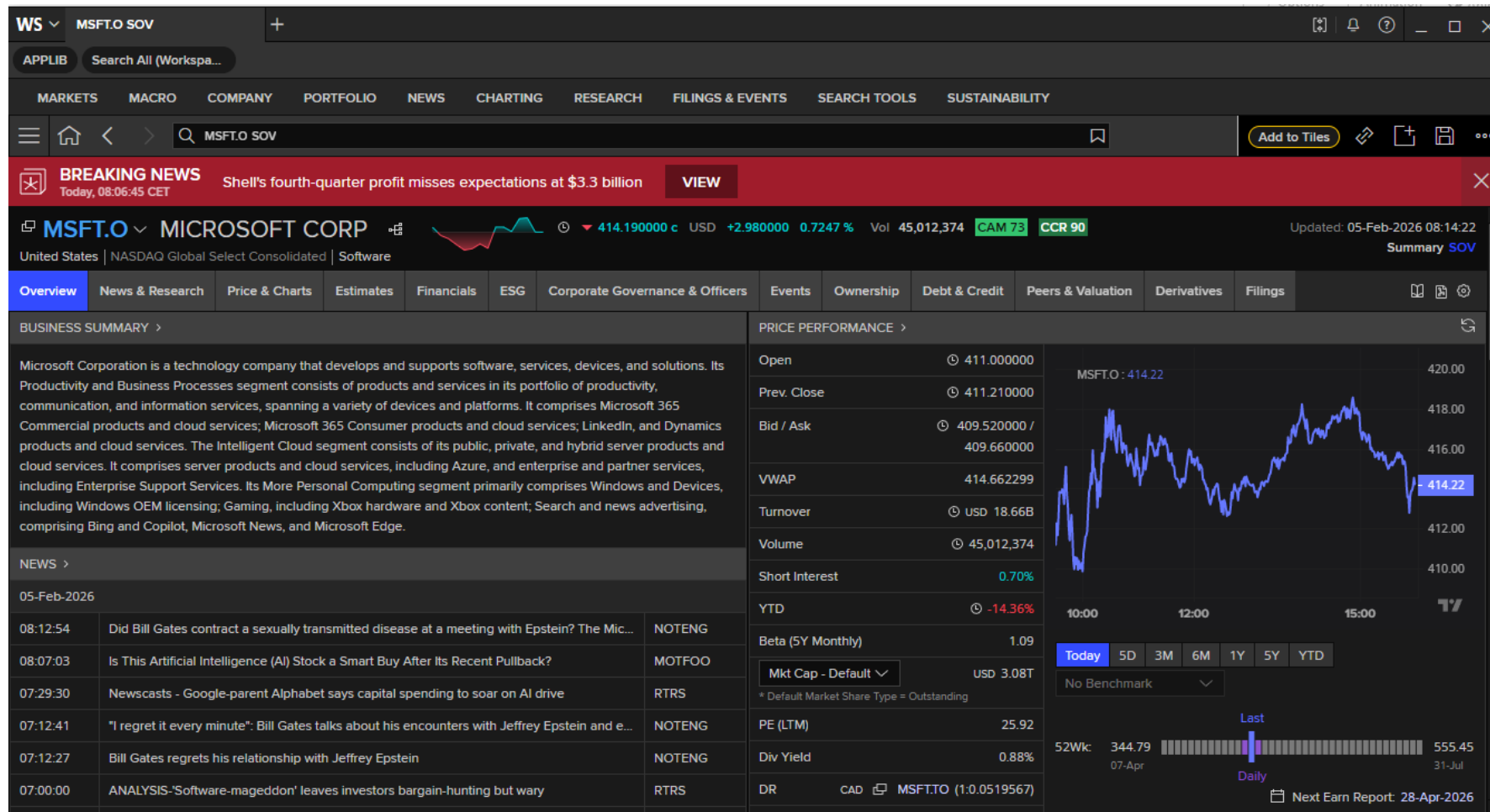
Hence, **collect and save the data** for later running the analyses

1. Refinitiv/LSEG

Available data:

- Accounting (book value, sales, capex, R&D, earnings)
- Financials (bond, stock, option)
- Forecasts (EPS, BPS, price)
- Ownership
- ESG (data points, scores, controversies)
- Filings (annual reports, earning calls, integrated reports)
- Fund information
- Country information
- News flow
- Prices (commodities, FOREX)
- M&A, Deals of PE

1. Refinitiv/LSEG



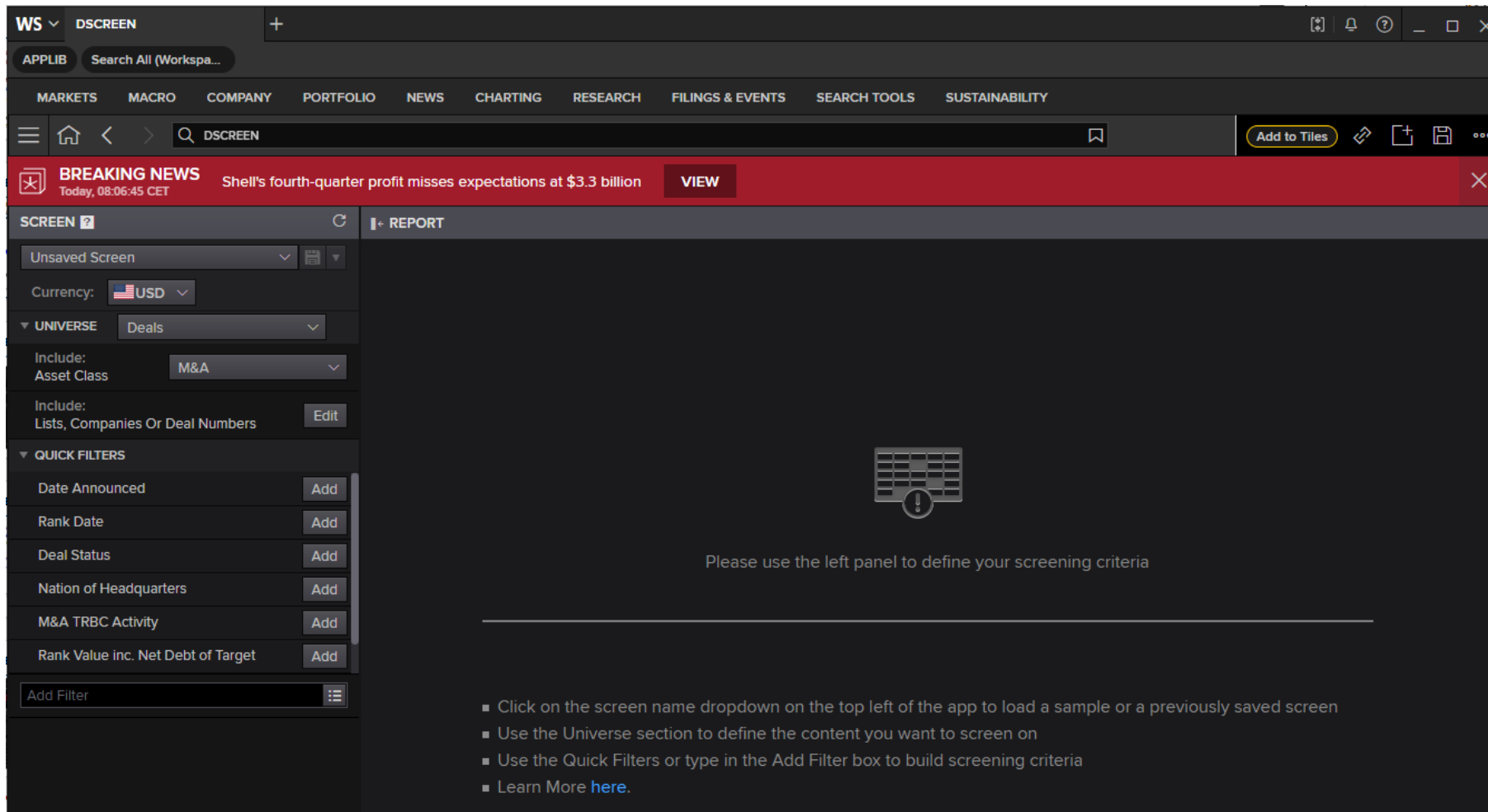
We search for **Microsoft**

We type Microsoft in the search bar or use its **RIC** (security identifier)

Also works with *ISIN* or *CUSIP*, other security identifiers

Let us explore the content together

1. Refinitiv/LSEG



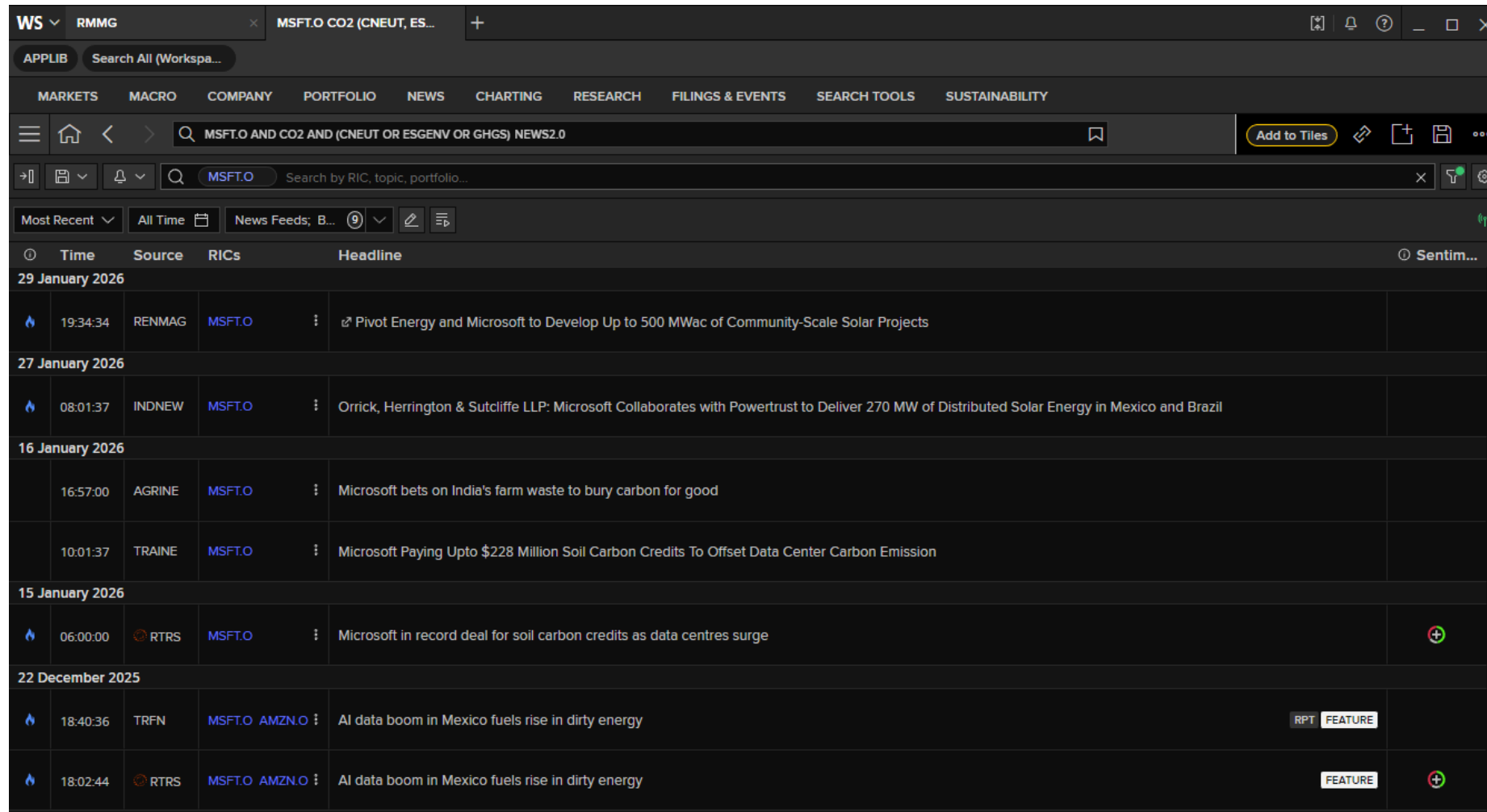
Screensers

Companies

Deals screener (bond,
stock, M&A, PE)

Fund screener

1. Refinitiv/LSEG



The screenshot displays the Refinitiv/LSEG news monitor interface. The top navigation bar includes tabs for APPLIB, Search All (Workspa...), and various market-related categories like MARKETS, MACRO, COMPANY, PORTFOLIO, NEWS, CHARTING, RESEARCH, FILINGS & EVENTS, SEARCH TOOLS, and SUSTAINABILITY. The search bar contains the query "MSFT.O AND CO2 AND (CNEUT OR ESGENV OR GHGS) NEWS2.0". Below the search bar, there are filters for "Most Recent" and "All Time", and a "News Feeds; B..." dropdown. The main table lists news articles with columns for Time, Source, RICs, and Headline. The articles are grouped by date, starting from 29 January 2026 and going back to 22 December 2025. The interface also includes a "Sentim..." column and a "FEATURE" button for some articles.

	Time	Source	RICs	Headline	Sentim...
29 January 2026					
🔥	19:34:34	RENMA	MSFT.O	🔗 Pivot Energy and Microsoft to Develop Up to 500 MWac of Community-Scale Solar Projects	
27 January 2026					
🔥	08:01:37	INDNEW	MSFT.O	Orrick, Herrington & Sutcliffe LLP: Microsoft Collaborates with Powertrust to Deliver 270 MW of Distributed Solar Energy in Mexico and Brazil	
16 January 2026					
	16:57:00	AGRINE	MSFT.O	Microsoft bets on India's farm waste to bury carbon for good	
	10:01:37	TRAINE	MSFT.O	Microsoft Paying Up to \$228 Million Soil Carbon Credits To Offset Data Center Carbon Emission	
15 January 2026					
🔥	06:00:00	RTSR	MSFT.O	Microsoft in record deal for soil carbon credits as data centres surge	+
22 December 2025					
🔥	18:40:36	TRFN	MSFT.O AMZN.O	AI data boom in Mexico fuels rise in dirty energy	RPT FEATURE
🔥	18:02:44	RTSR	MSFT.O AMZN.O	AI data boom in Mexico fuels rise in dirty energy	FEATURE +

News monitor

1. Refinitiv/LSEG

WS ▾ DIB

APPLIB Search All (Workspa...

MARKETS MACRO COMPANY PORTFOLIO NEWS CHARTING RESEARCH FILINGS & EVENTS SEARCH TOOLS SUSTAINABILITY

⌵ ⌵ ⌵ ⌵ ⌵ DIB

Add an instrument Remove Instrument

earnings Recently Used All Items

Data Item Name	Data Item Code
Earnings	EARNINGS
Earnings	TR.SACTTarEarningsBeforeAnn
Earnings Per Share - Mean	TR.EPSMean
Earnings Per Share Reported - Mean	TR.RepEPSMean
Preferred Earnings - Mean Estimate	TR.PreferredMeasureMeanEst
Earnings Per Share - SmartEstimate®	TR.EpsSmartEst
Earnings Per Share - Predicted Surprise PCT	TR.EpsPreSurprisePct
Price / EPS (SmartEstimate®)	TR.FwdPtoEPSSmartEst
EBIT Margin - %	TR.F.EBITMargPct
EBITDA Margin - %	TR.F.EBITDAMargPct
Income before Taxes to EBIT	TR.F.IncBefTaxtoEBIT
Equity Earnings/(Loss) before Taxes including Non-Recurring	TR.F.EqEarnLossBefTaxInclNonRecur
Normalized Pre-tax Profit	TR.F.NormPretaxProf
Income before Taxes	TR.F.IncBefTax
Net Income after Tax	TR.F.NetIncAfterTax
Equity in Earnings/(Loss) of Affiliates (after Taxes)	TR.F.EqInEarnLossOfAffilAfterTax

Description

Earnings

EARNINGS

Definition:
Latest reported earnings per share.
FieldID: 34

Show only: ☒ Cross-Asset Items ☐ Favorites Sort By: ☒ Relevance ☐ A - Z

EARNINGS

Data Item Browser
- DIB -

1. Refinitiv/LSEG

The screenshot displays the Refinitiv CODECR (Code Generator) interface. At the top, the workspace is named 'WS' and 'CODECR'. A search bar at the top left contains 'Search All (Workspa...'. Below this, a navigation bar includes tabs for MARKETS, MACRO, COMPANY, PORTFOLIO, NEWS, CHARTING, RESEARCH, FILINGS & EVENTS, SEARCH TOOLS, and SUSTAINABILITY. The main interface is divided into several sections:

- Instruments:** A search bar with 'MSFT.O' entered.
- DATA ITEMS:** A search bar with 'Search Data Items' and a list of categories on the left. The 'Number of Shares' item is selected, showing its description: 'TR.FundNumberOfShares' and 'Definition: Number of shares held. Category: Fund Holdings'.
- Selected Items:** A list containing 'TR.FundNumberOfShares' with a 'Drag to Reorder' option and a 'Clear All' button.
- OUTPUT:** A section with a 'Python' tab showing the generated code:

```
import refinitiv.data as rd
rd.open_session()
df = rd.get_data(
    universe = ['MSFT.O'],
    fields = ['TR.FundNumberOfShares']
)
display(df)
```

**Python Code
Generator
- CODECR -**

2. Collecting data with Python

- Why **Python**?
- **Anaconda** bundles Python, Jupyter notebooks, and common data packages in one installer.
 - Installs Python automatically
 - Works on both Mac and Windows
- We download and install it : <http://anaconda.com/download>



2. Collecting data with Python

Anaconda Navigator

File Help

ANACONDA.NAVIGATOR

Update Now Connect

Home Environments Learning Community

All applications on base (root) Channels

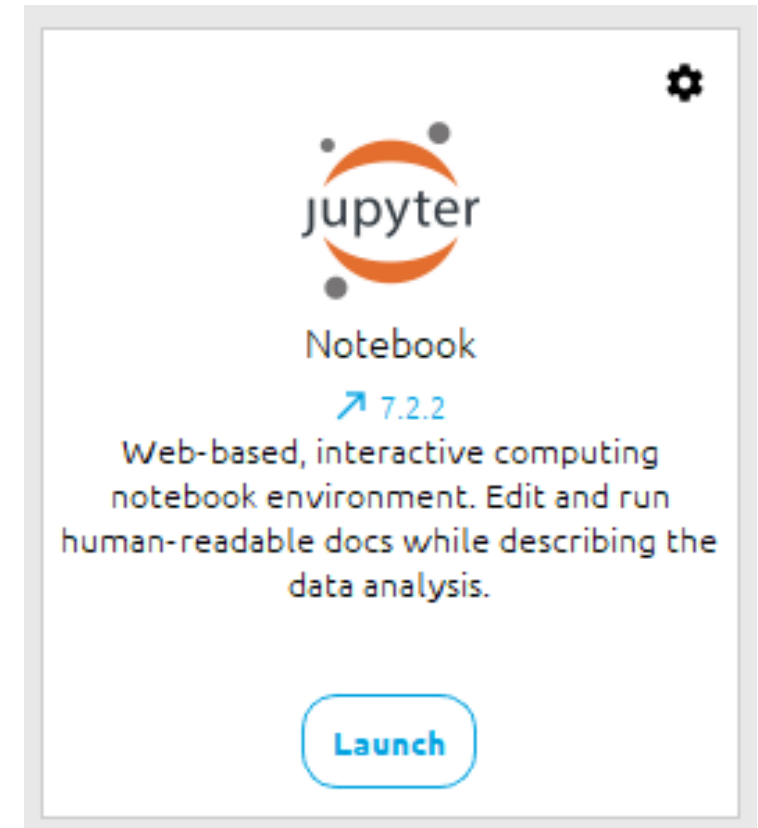
 PyCharm The only Python IDE you need – built for data and AI/ML professionals. Supercharged with an AI-enhanced IDE experience. Free forever, plus one month of Pro included. Install	 Anaconda AI Navigator Access various large language models (LLMs) curated by Anaconda, and start leveraging secure local AI today. Install	 Anaconda Toolbox 4.40.0 Anaconda Assistant JupyterLab supercharged with a suite of Anaconda extensions, starting with the Anaconda Assistant AI chatbot. Install	 Anaconda Cloud Notebooks Cloud-hosted notebook service from Anaconda. Launch a preconfigured environment with hundreds of packages and store project files with persistent cloud storage. Launch	 Qt Console 5.6.1 PyQt GUI that supports inline figures, proper multiline editing with syntax highlighting, graphical calltips, and more. Launch	 Spyder 6.0.7 Scientific Python Development Environment. Powerful Python IDE with advanced editing, interactive testing, debugging and introspection features. Launch
 EduBlocks Web-based coding platform from Anaconda designed for students. Learn Python coding through an interactive, block-based visual environment. Launch	 IBM watsonx IBM watsonx is an enterprise-ready AI platform including a data store, model builder, and AI model management and monitoring. Launch	 Oracle Cloud Infrastructure Oracle Data Science Service OCI Data Science offers a machine learning platform to build, train, manage, and deploy your machine learning models on the cloud with your favorite open-source tools. Launch	 PyScript Code and share Python in the Browser. A vibrant community of makers, builders, and hackers building the next frontier of Python-powered web applications. Launch	 PythonAnywhere Host, run, and code Python in the cloud! Get started for free. Launch	 anaconda_powershell_prompt 1.1.0 Opens a PowerShell instance with conda activated (requires menuinst 2.1.1 or greater). Install
 anaconda_prompt 1.1.0 Opens a terminal instance with conda activated (requires menuinst 2.1.1 or greater). Launch	 CMD.exe Prompt 0.1.1 Run a cmd.exe terminal with your current environment from Navigator activated. Launch	 console_shortcut_miniconda 0.1.1 Anaconda Powershell Prompt Launch	 Glueviz 1.4.0 Multidimensional data visualization across files. Explore relationships within and across files. Launch	 JupyterLab 4.5.3 An extensible environment for interactive and reproducible computing, based on the Jupyter Notebook and built on top of the Jupyter ecosystem. Launch	 Jupyter Notebook 7.5.0 Web-based, interactive computing notebook environment. Edit and run code, visualize outputs, and describe the workflow. Launch

Anaconda Quick Start Environments
Jump into pre-configured environments by project or industry. Clean dependencies, faster development.
[Launch Your Environment](#)

Documentation
Anaconda Blog

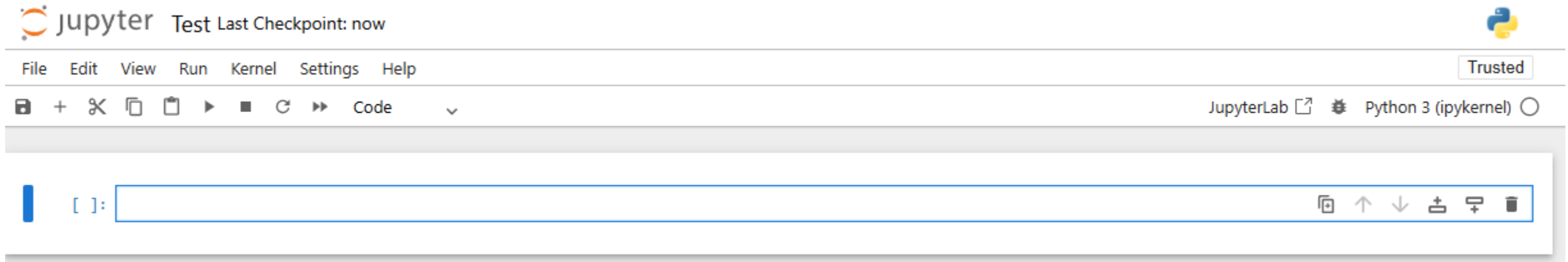
2. Collecting data with Python

- Install/launch **Jupyter Notebook**
- Jupyter Notebook is a tool that lets you write code, run it, and see the results all in one place—inside your web browser.
- Allows to run code step by step/ see output immediately (numbers, tables, plots) / add text explanations next to the code.



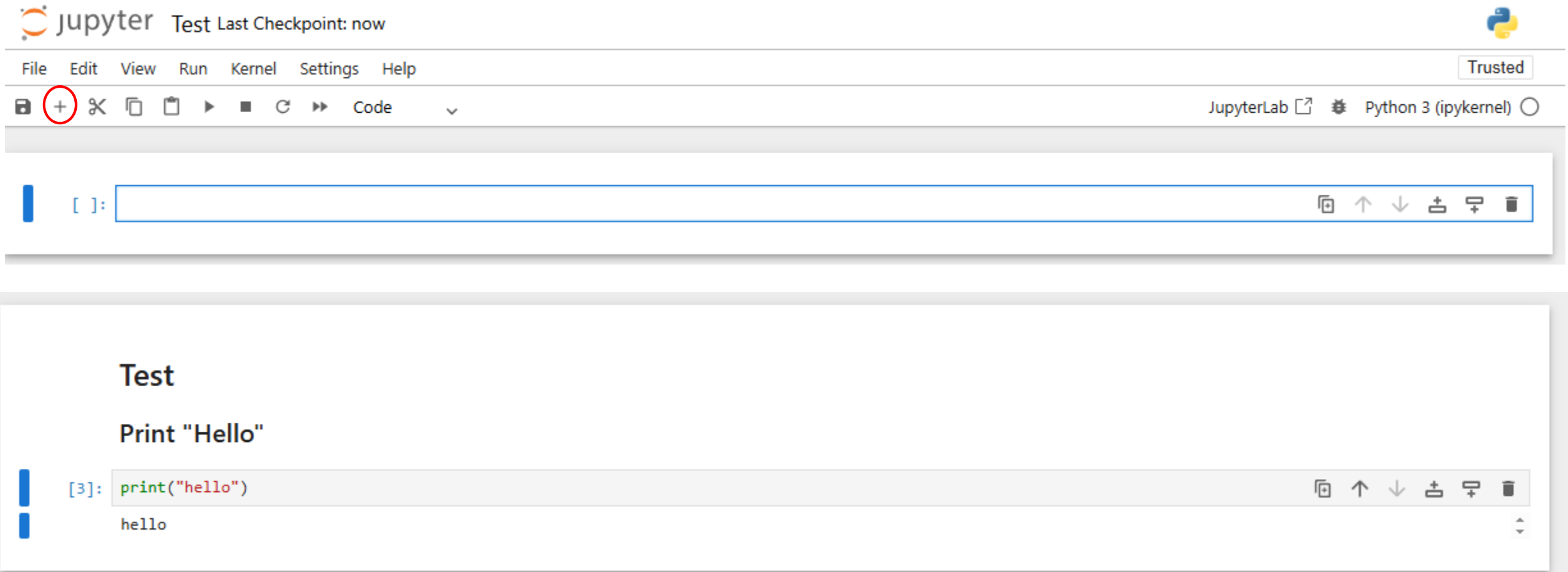
2. Collecting data with Python

- Launch Jupyter Notebook
- Choose folder
- File > New > Notebook (Python 3)
- Save it as *Test.ipynb*



2. Collecting data with Python

➤ With the + button, we can add text (markdown) or code section (python), and then execute



The screenshot displays the JupyterLab web interface. At the top, the 'jupyter' logo is on the left, and the status 'Test Last Checkpoint: now' is in the center. On the right, there is a Python logo and a 'Trusted' badge. Below this is a menu bar with 'File', 'Edit', 'View', 'Run', 'Kernel', 'Settings', and 'Help'. A toolbar follows, containing icons for saving, adding a new cell (highlighted with a red circle), deleting, copying, pasting, running, and other actions. The 'Code' dropdown menu is open. To the right of the toolbar, it says 'JupyterLab' with an external link icon, a settings gear icon, and 'Python 3 (ipykernel)' with a refresh icon. The main workspace shows a single code cell with the prompt '[]:' and an empty text input field. Below this, a second code cell is shown with the prompt '[3]:' and the code `print("hello")`. The output of this cell is 'hello'.

jupyter Test Last Checkpoint: now

File Edit View Run Kernel Settings Help

Trusted

JupyterLab Python 3 (ipykernel)

[]:

Test

Print "Hello"

[3]: `print("hello")`

hello

2. Collecting data with Python

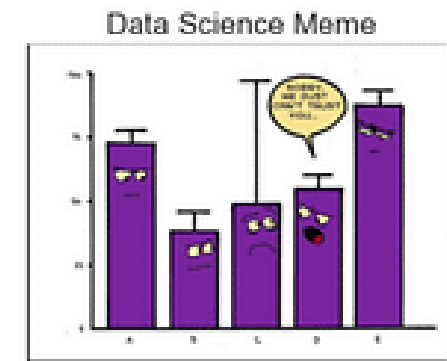
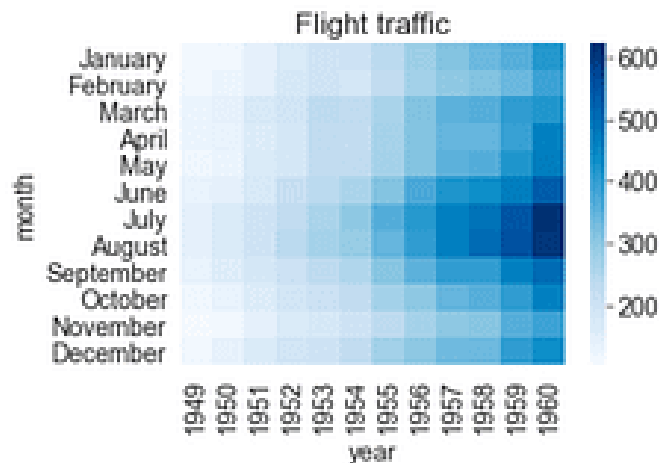
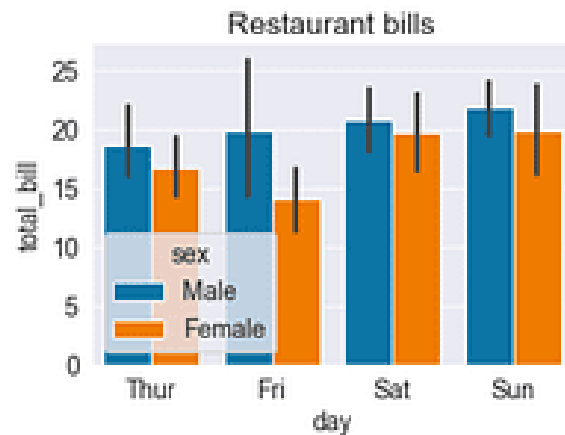
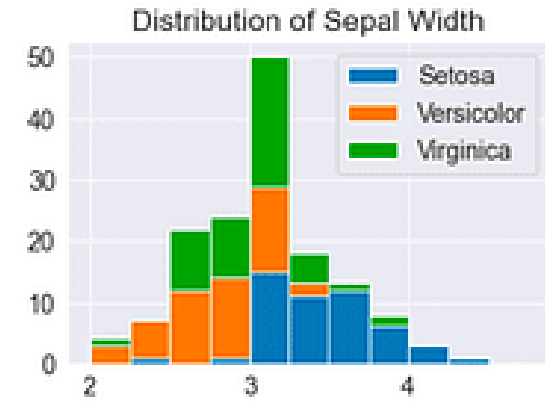
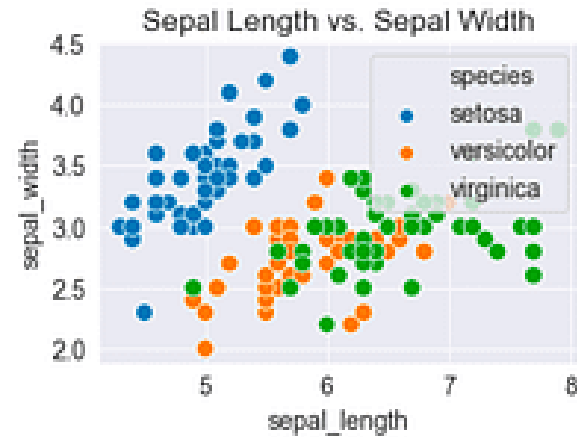
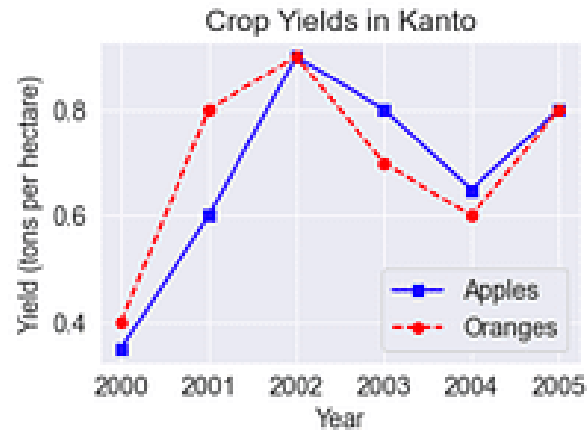
We switch to the **research_method_notebook**

All the material, including this presentation is available on **Blackboard**.

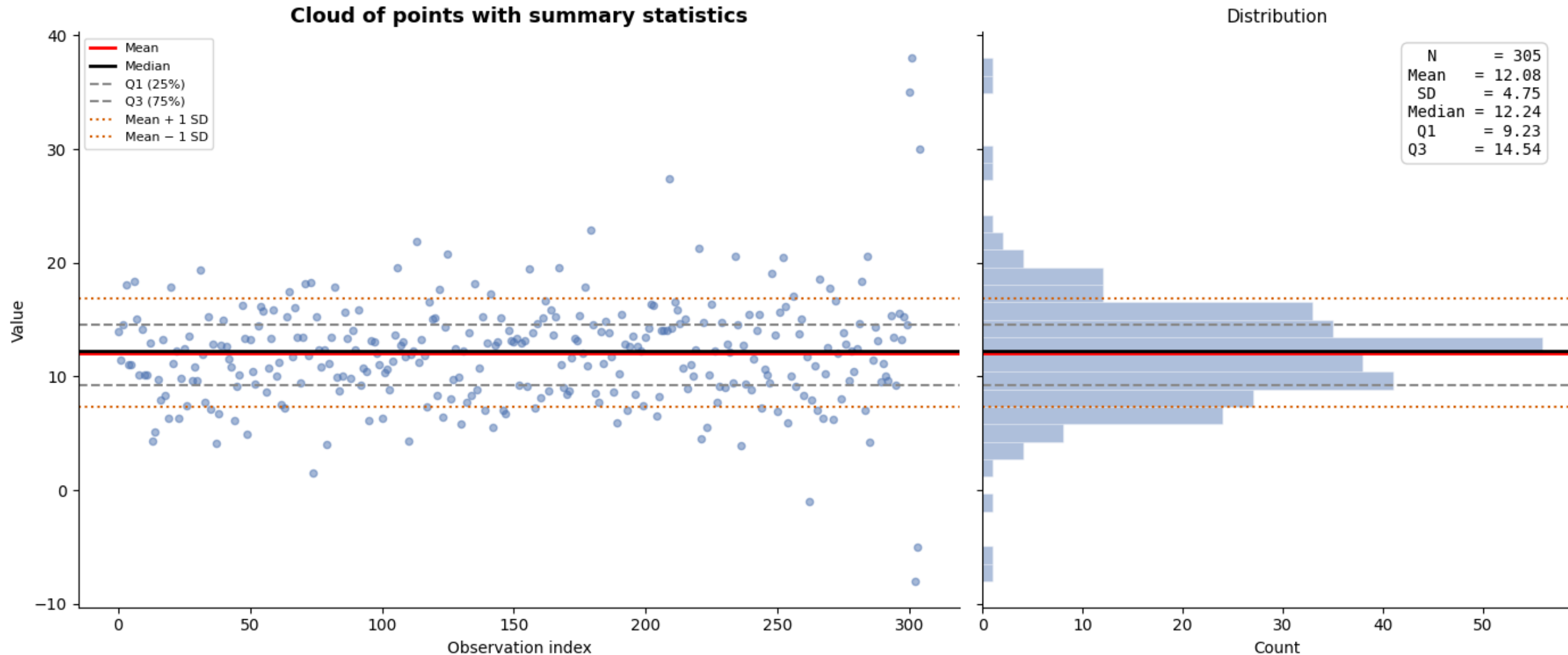
2. Collecting data with Python

Now that we have reviewed the part of the **research_method_notebook** dedicated to data collection, try to collect datasets possibly relevant for your own master thesis...

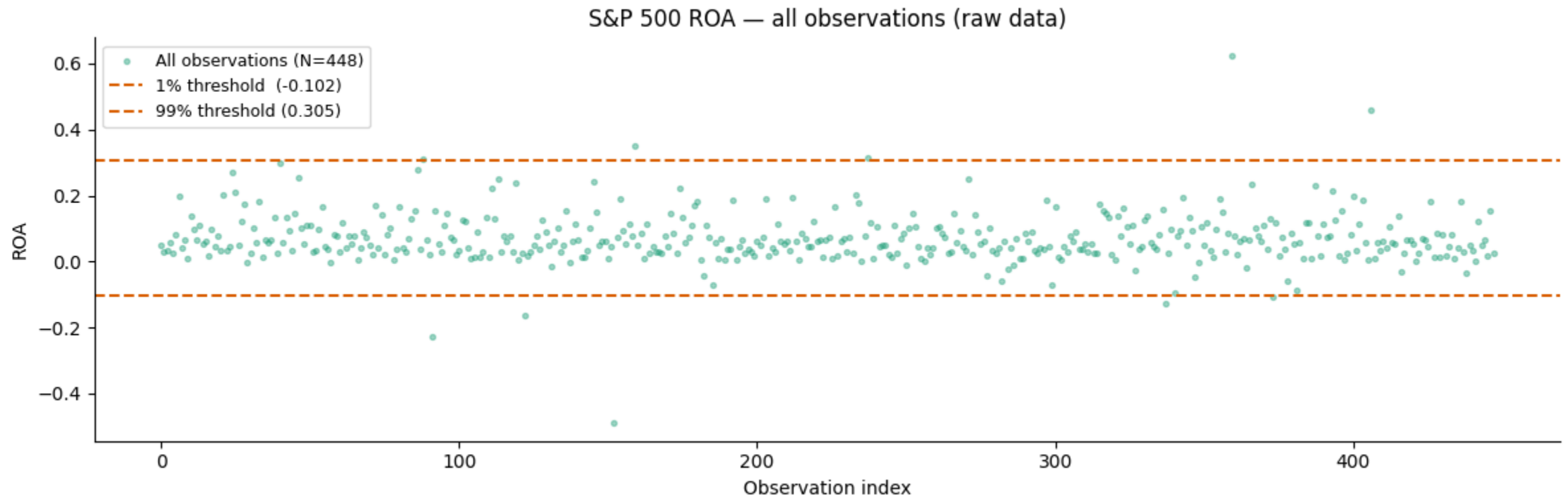
3. Data visualization in Python



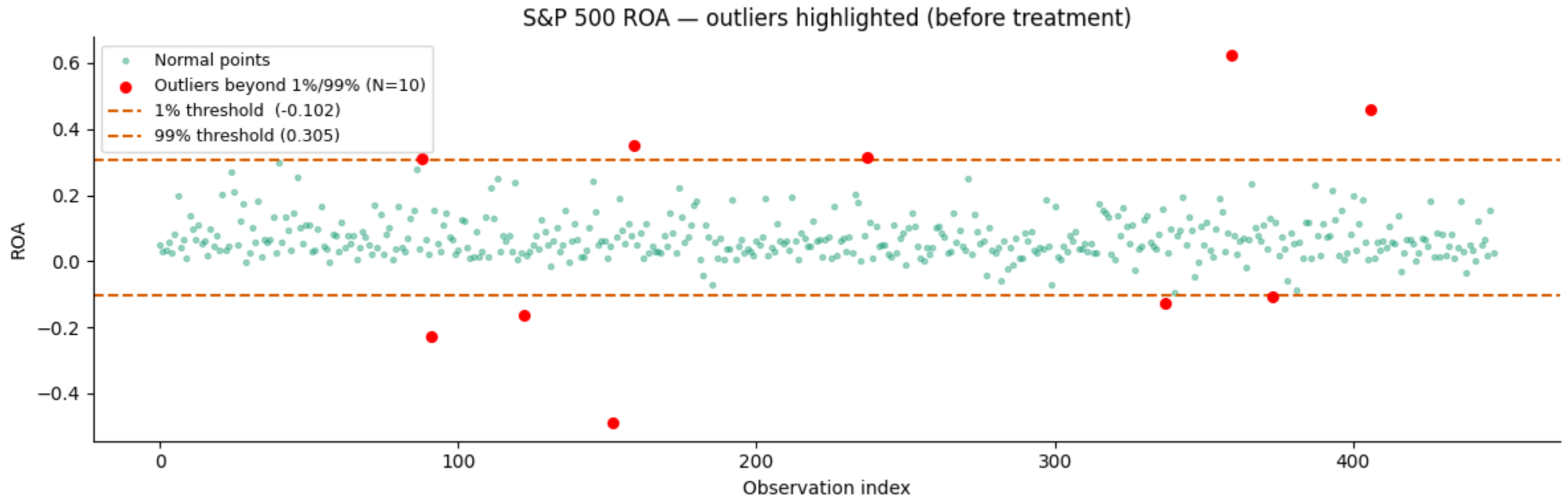
4. Descriptive statistics with Python



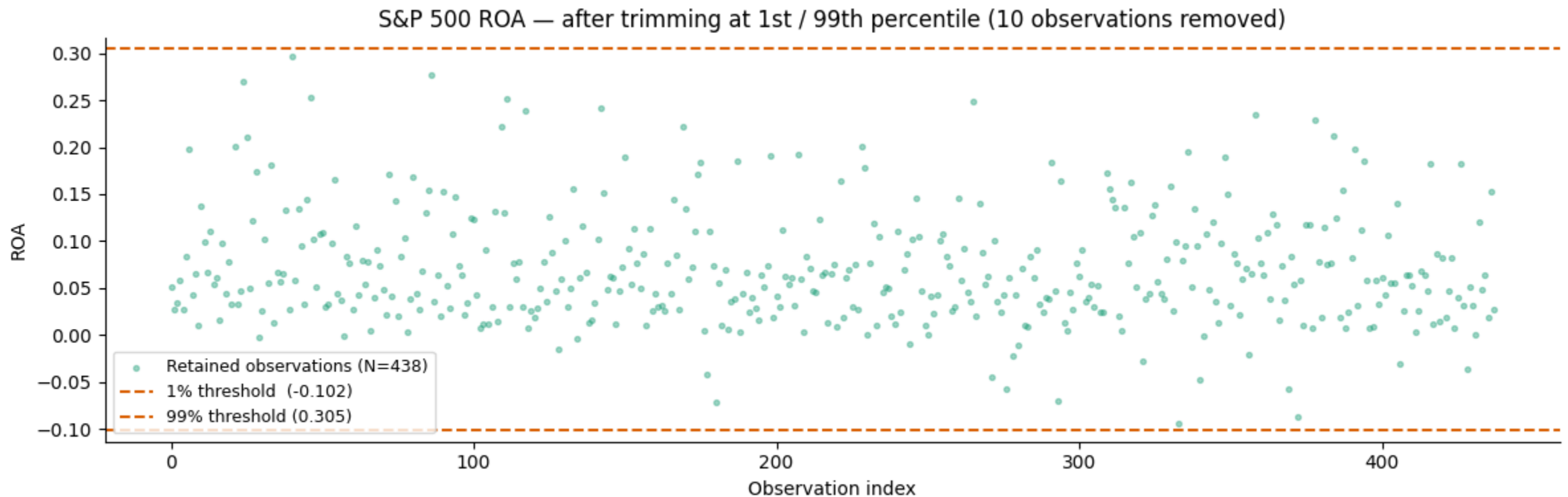
4. Winsorizing/Trimming



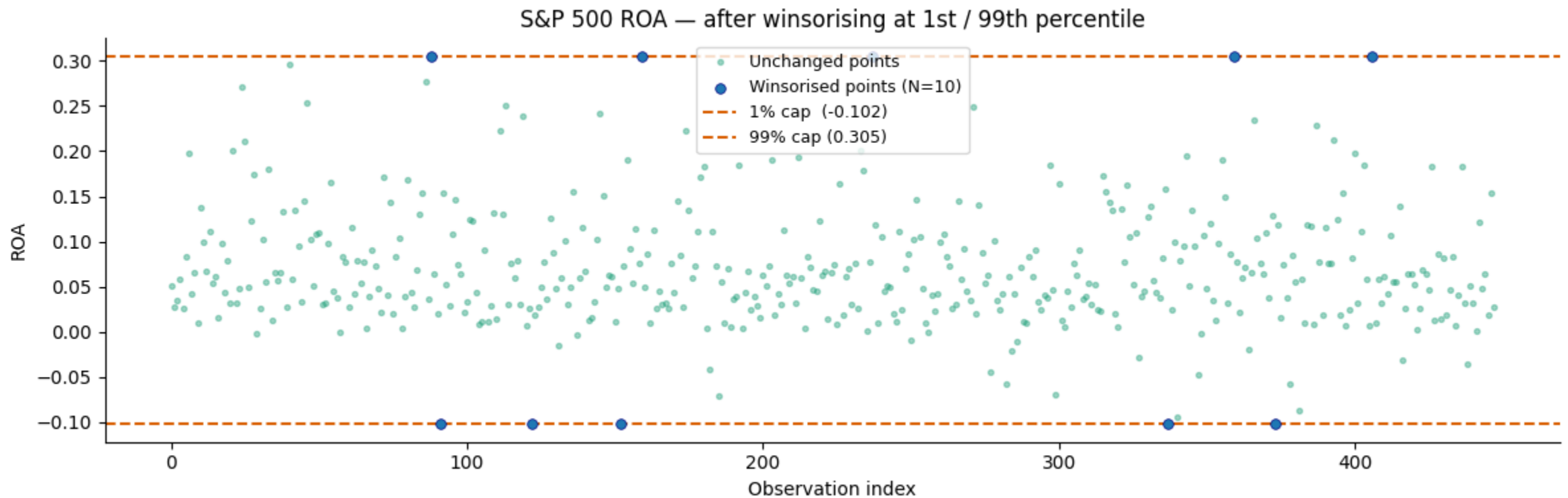
4. Winsorizing/Trimming



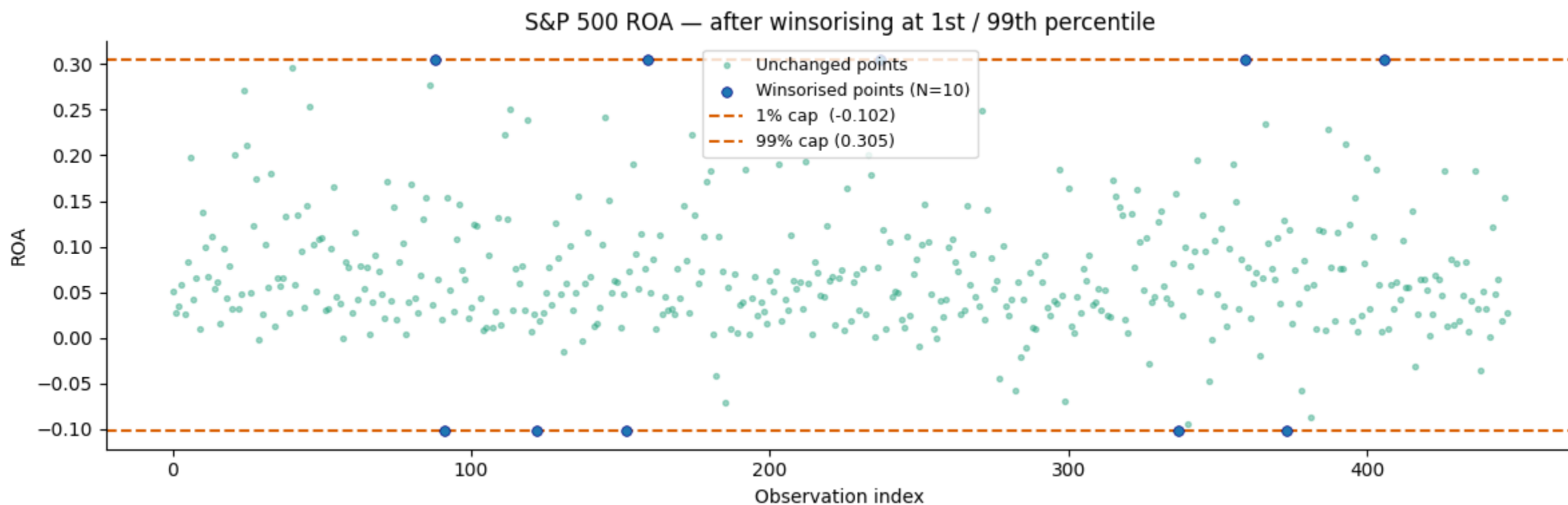
4. Winsorizing/Trimming



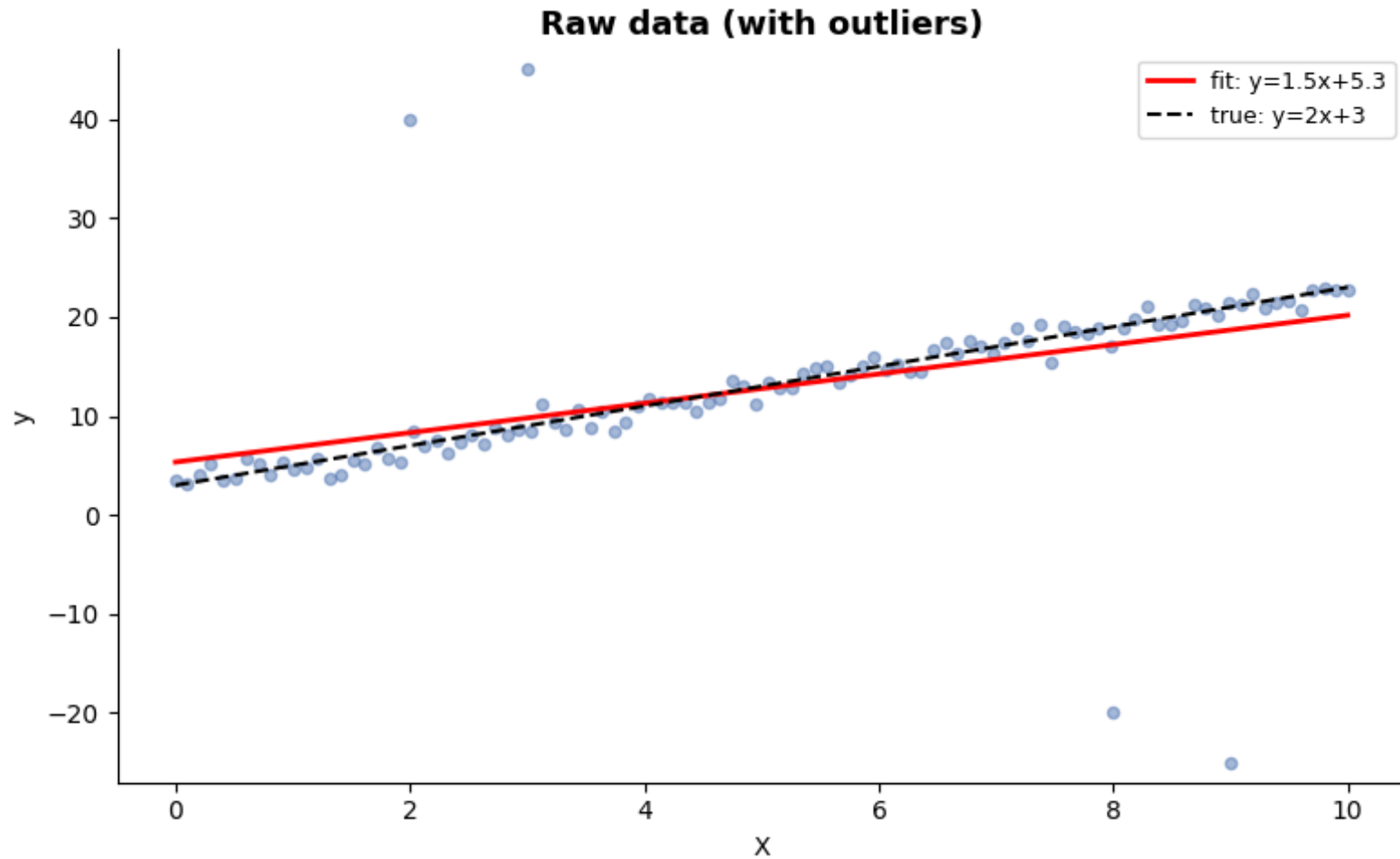
4. Winsorizing/Trimming



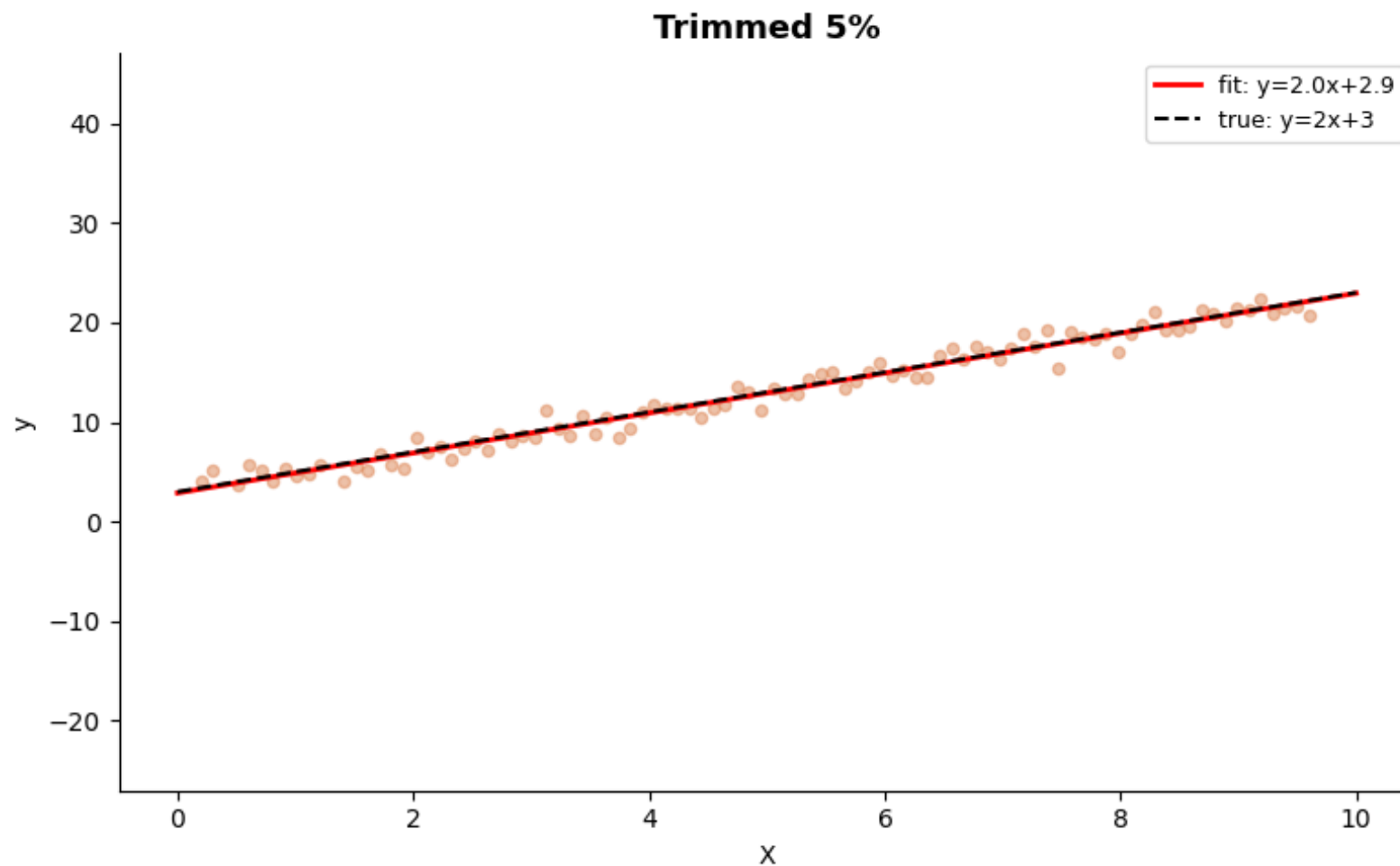
4. Winsorizing/Trimming



4. Winsorizing/Trimming



4. Winsorizing/Trimming



5. T-test

A **population** is the entire set of individuals, firms, or observations you want to draw conclusions about (e.g. all listed firms in a given year). It's the "truth" — but in practice it's usually too large or impossible to observe in full.

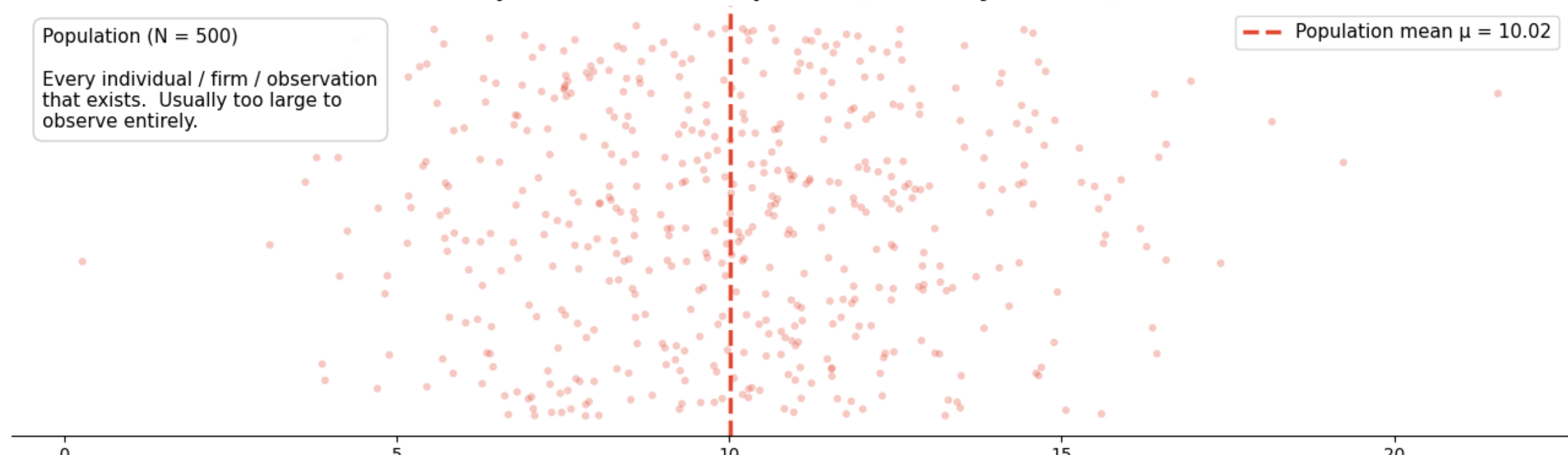
A **sample** is the subset you actually collect and analyse. Because it's only a piece of the population.

Population — the full picture (we rarely see this)

Population (N = 500)

Every individual / firm / observation that exists. Usually too large to observe entirely.

--- Population mean $\mu = 10.02$



5. T-test

A **population** is the entire set of individuals, firms, or observations you want to draw conclusions about (e.g. all listed firms in a given year). It's the "truth" — but in practice it's usually too large or impossible to observe in full.

A **sample** is the subset you actually collect and analyse. Because it's only a piece of the population.

Sample — the slice of reality we work with

Sample ($n = 40$)

The subset we actually observe. $\bar{x}$ estimates μ , but comes with uncertainty — it won't be exact.

Rest of population (460)
Our sample ($n = 40$)
Population mean $\mu = 10.02$ (unknown)
Sample mean $\bar{x} = 9.88$ (what we compute)

0

5

10

15

20

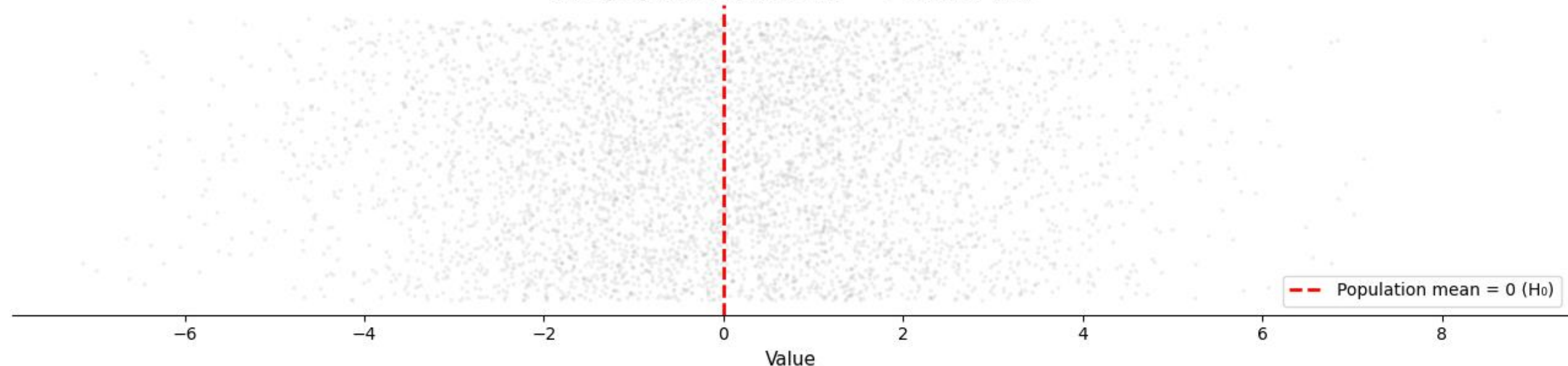
Value

5. T-test

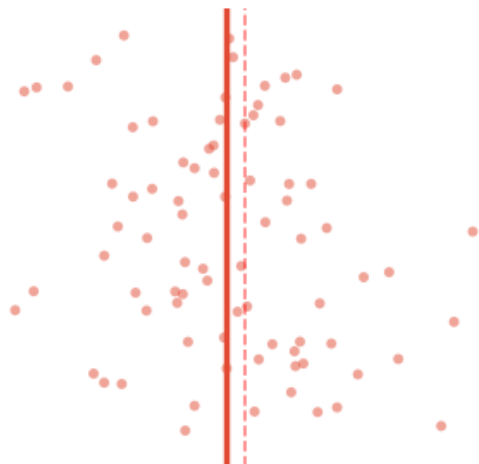
Null hypothesis that the mean is zero in the population (H_0). We generate a **p-value** to reject/not reject the hypothesis.

The **p-value** is the probability of observing a sample mean as extreme as ours (or more extreme) if H_0 were actually true.

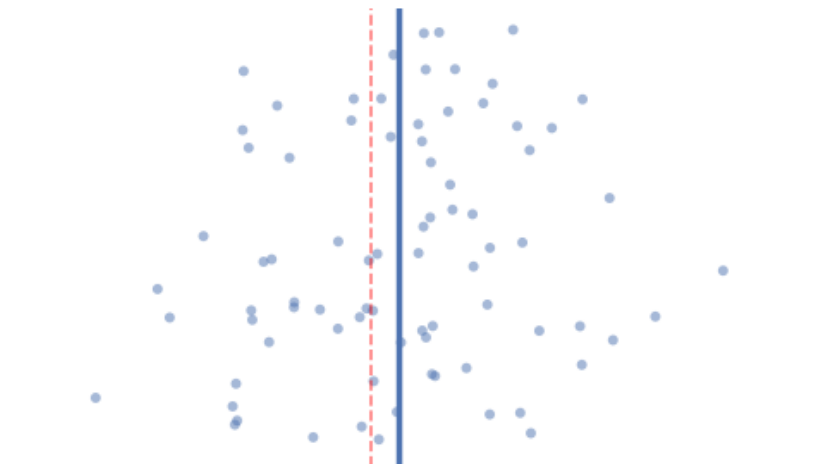
The population (mean = 0 under H_0)



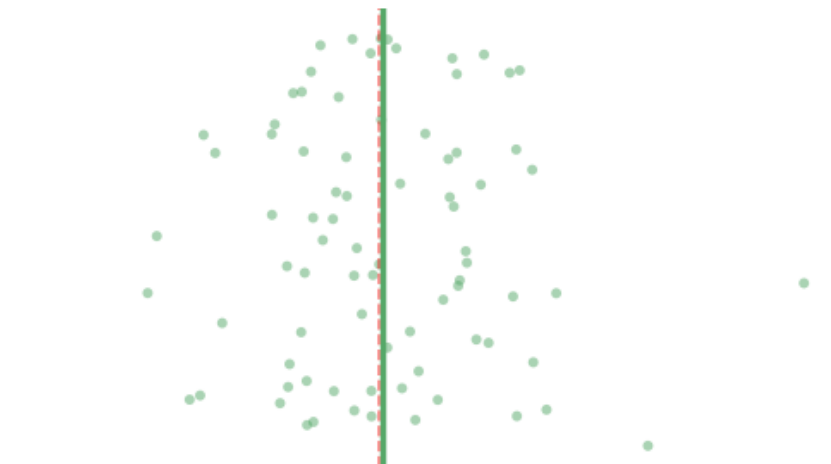
Draw #1: $\bar{x} = -0.35$



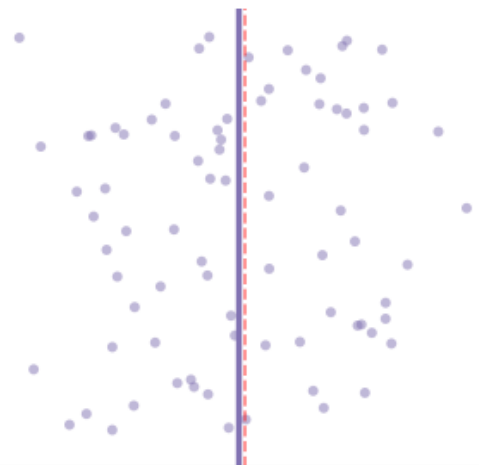
Draw #2: $\bar{x} = 0.57$



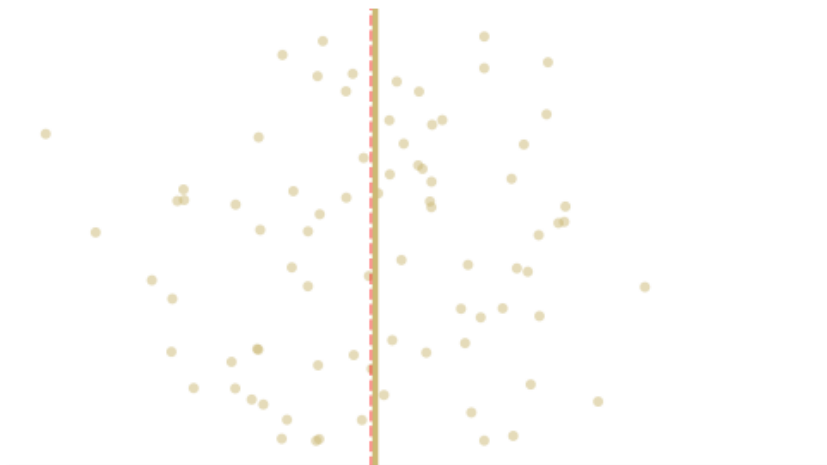
Draw #3: $\bar{x} = 0.09$



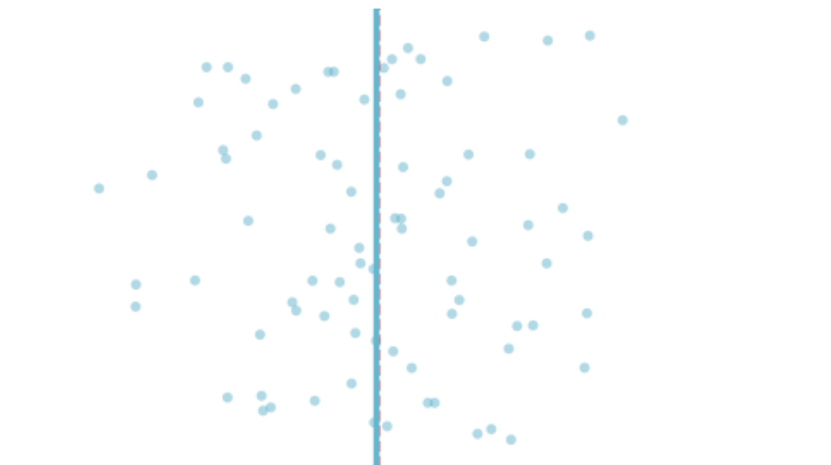
Draw #4: $\bar{x} = -0.13$



Draw #5: $\bar{x} = 0.08$

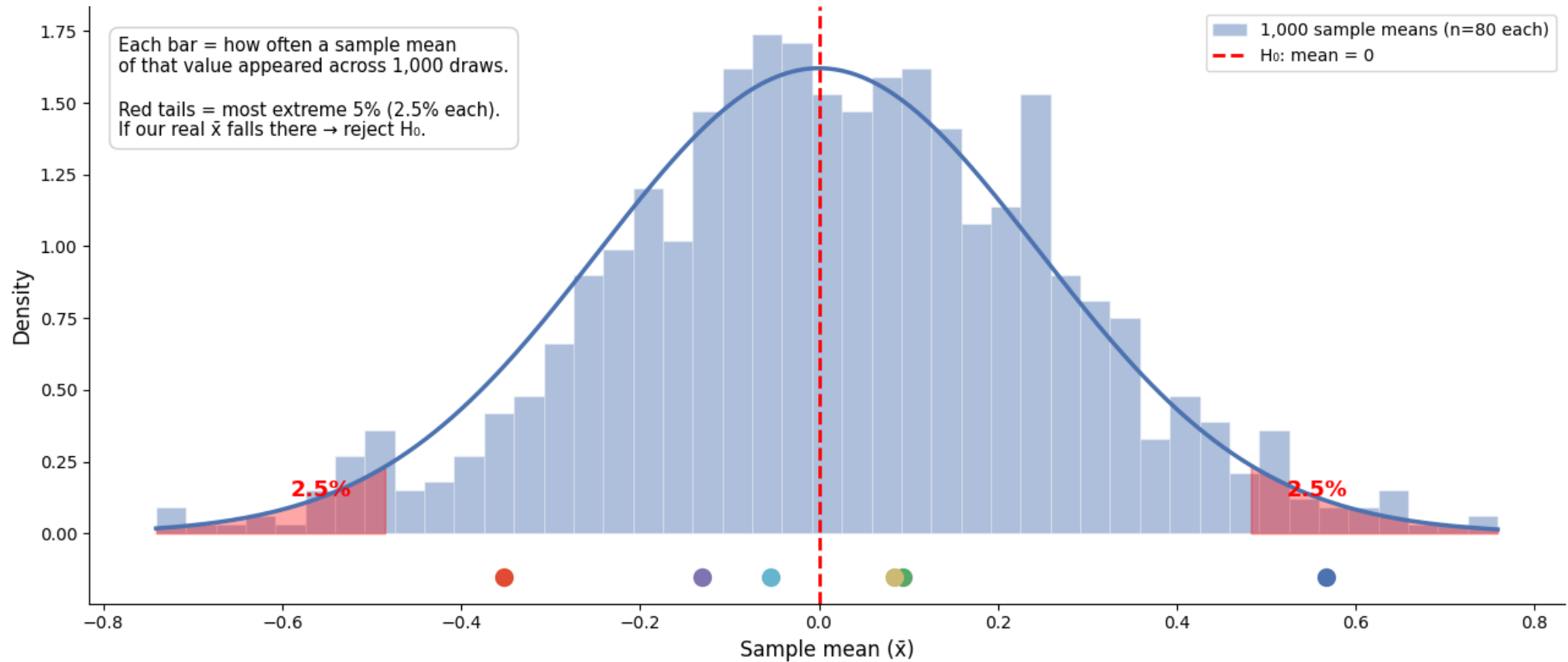
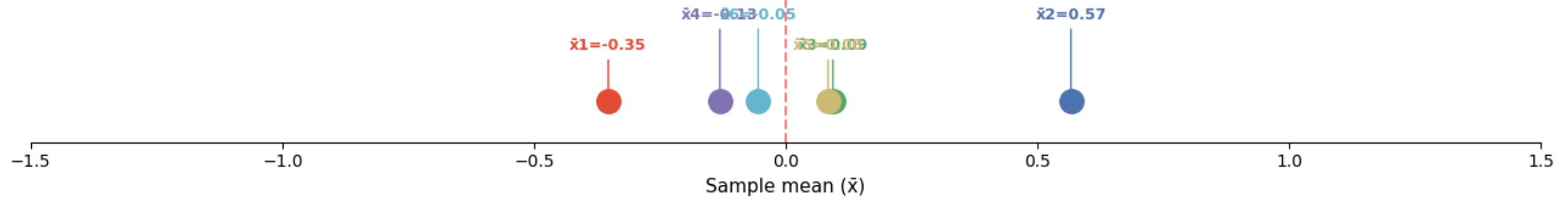


Draw #6: $\bar{x} = -0.05$

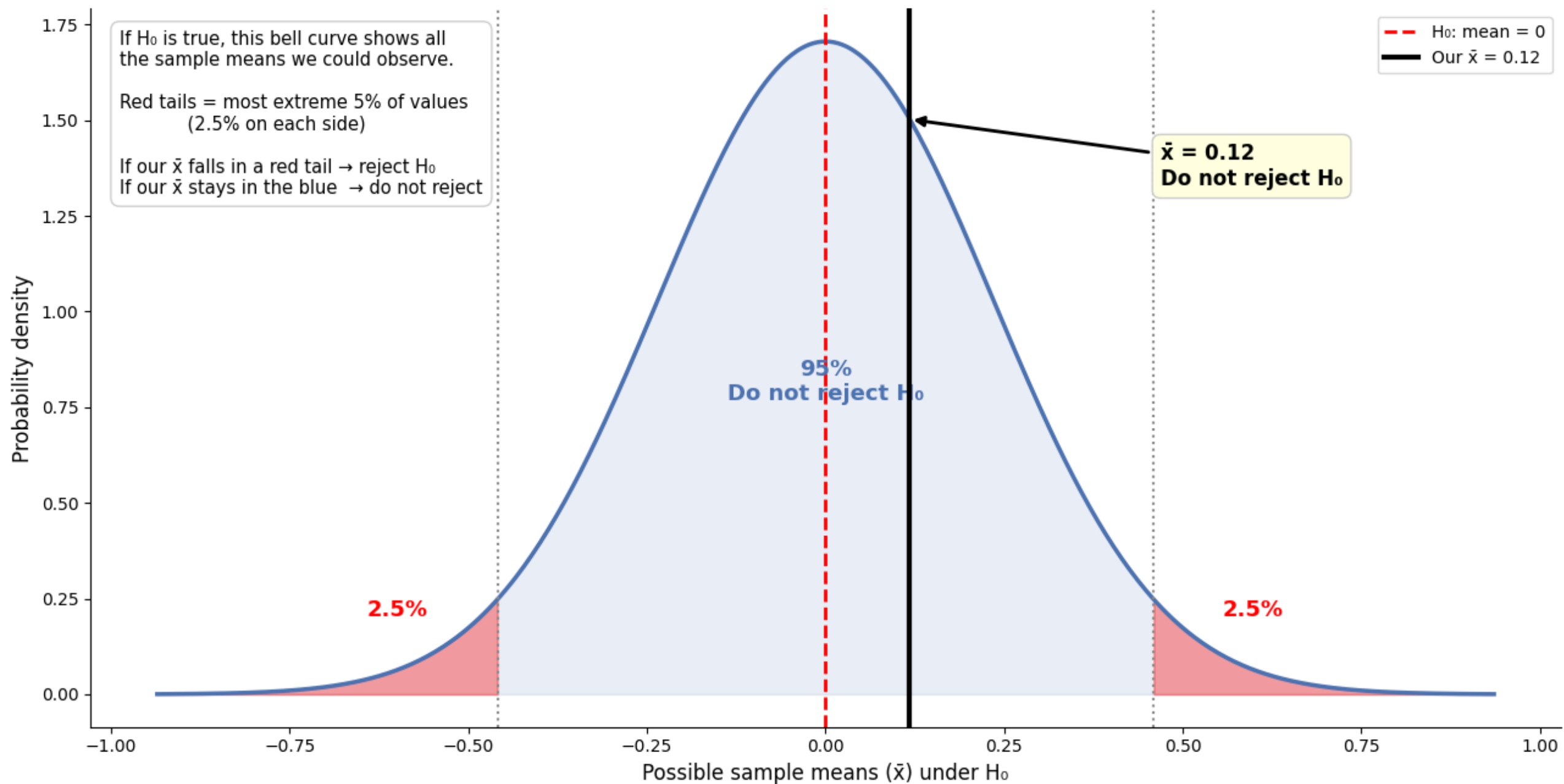


Value

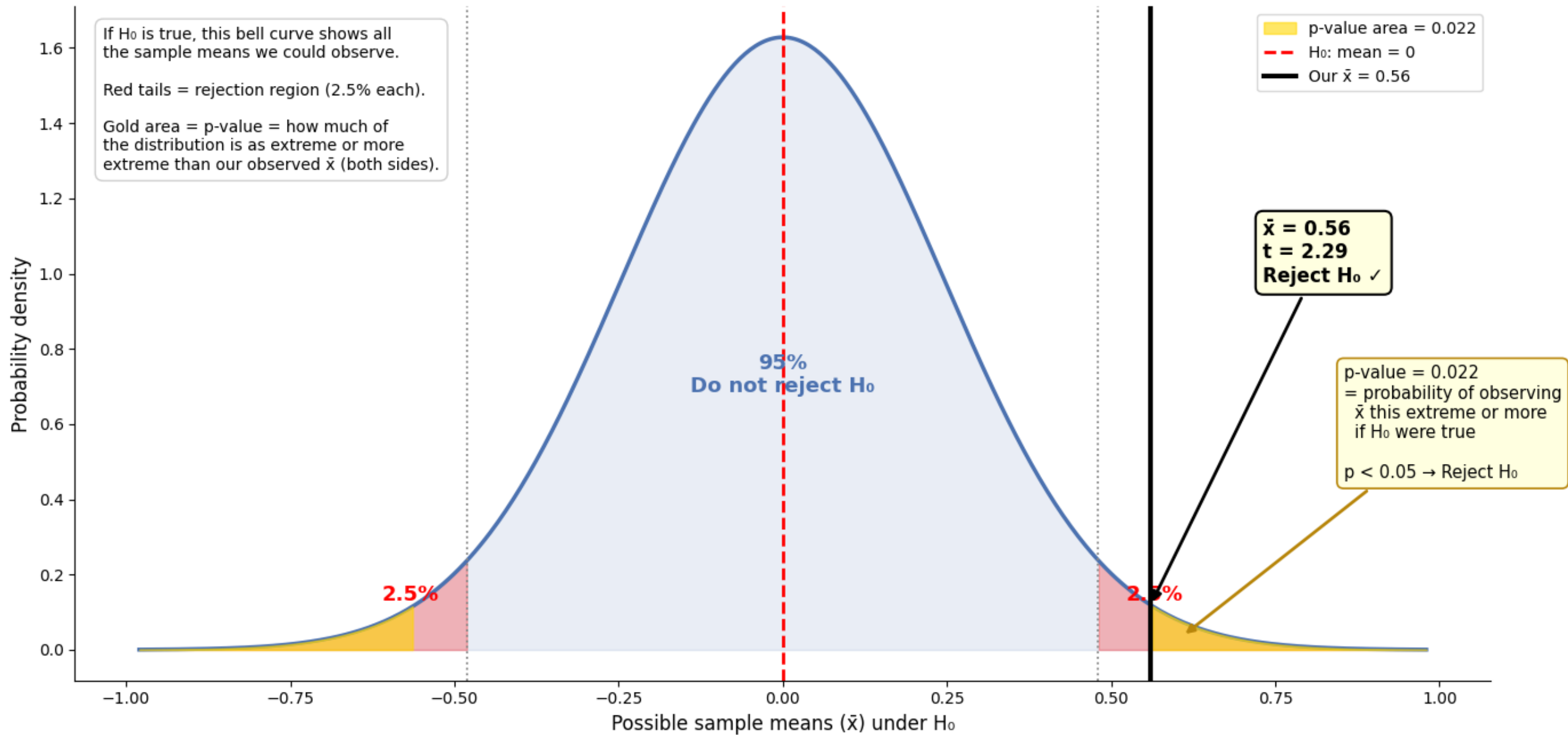
6 draws $\rightarrow$ 6 dots. What if we did this 1,000 times?



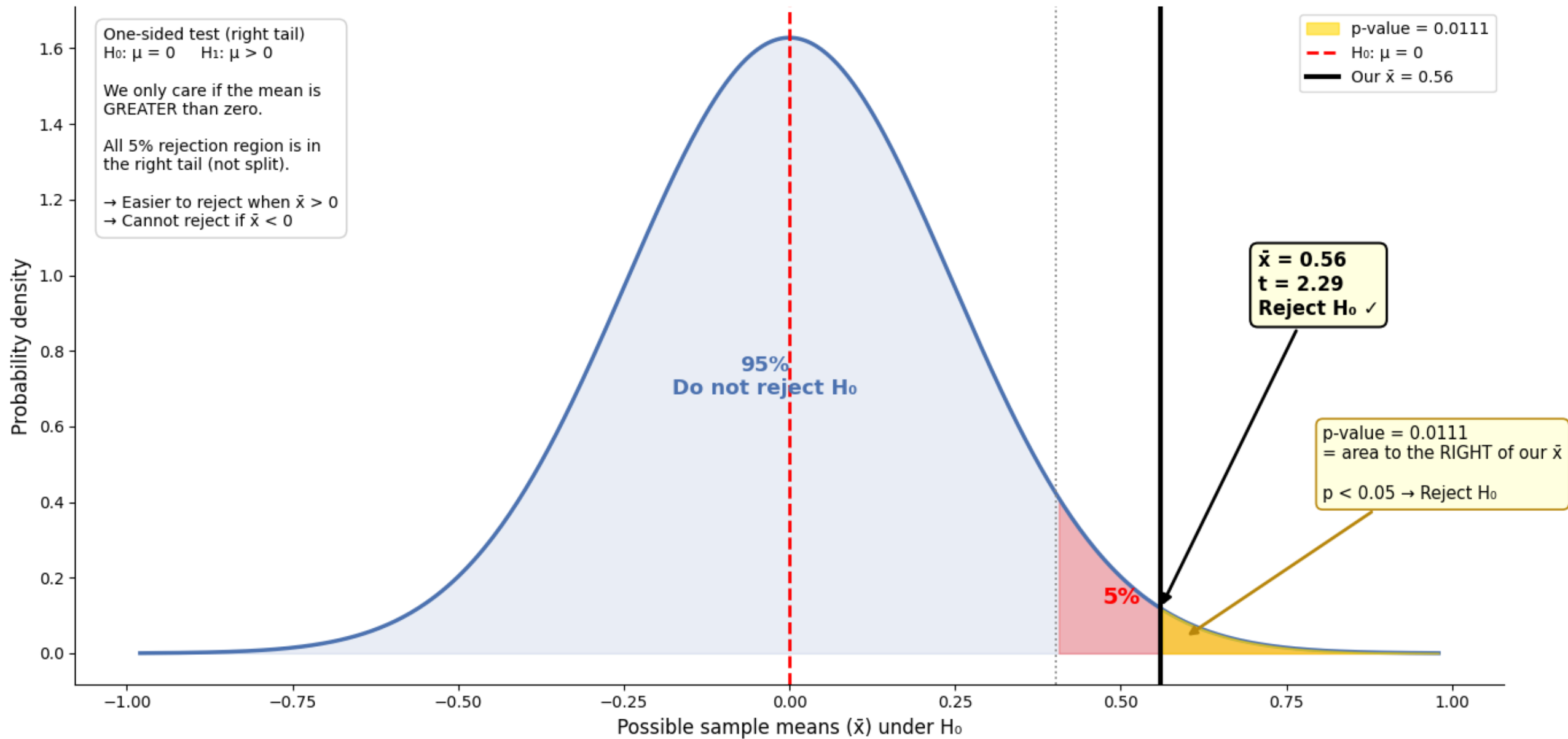
Reject H_0 if our sample mean falls in the extreme 5% (2.5% each tail)



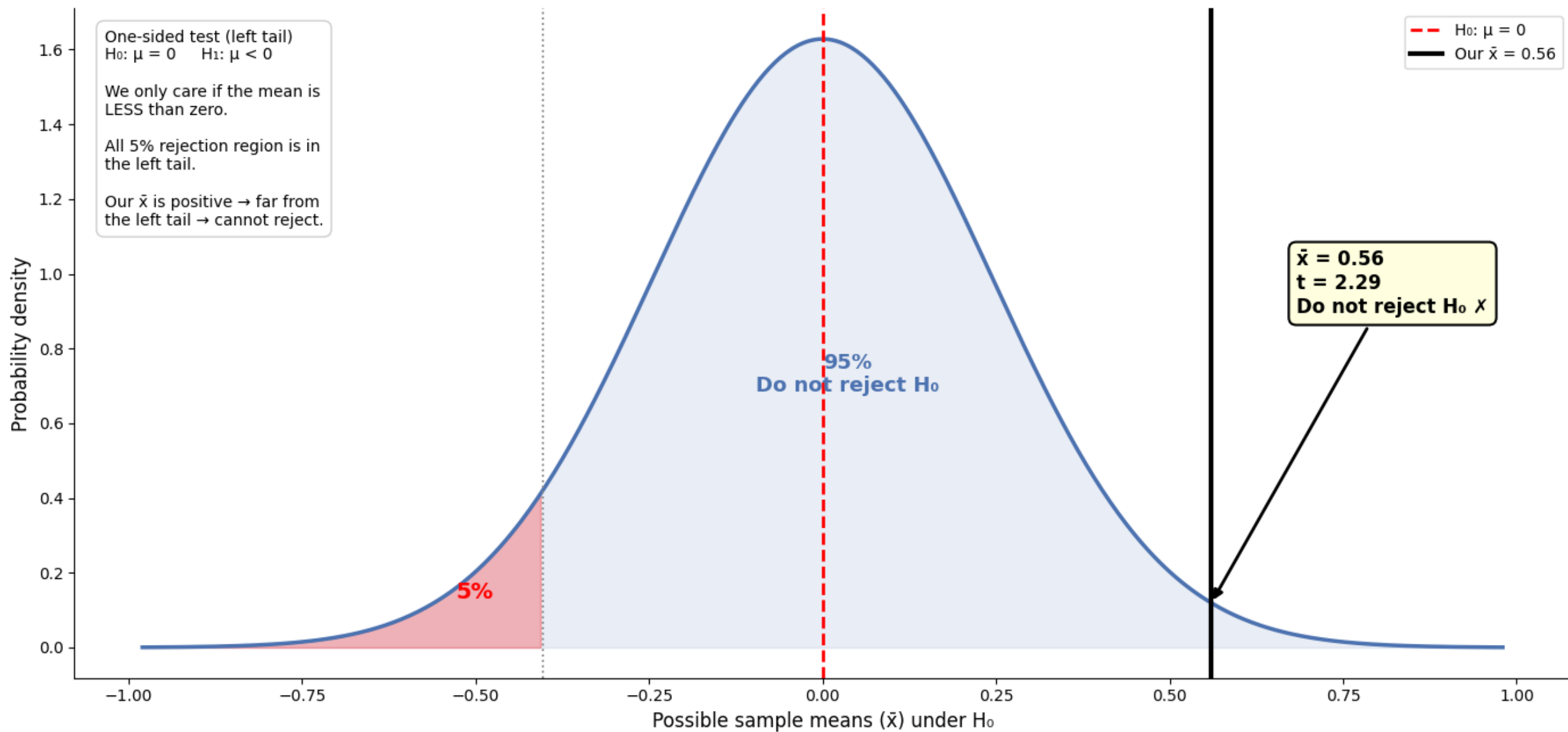
Reject H_0 if our sample mean falls in the extreme 5% (2.5% each tail)



One-sided test: $H_0: \mu = 0$ vs $H_1: \mu > 0$ (5% in right tail)



One-sided test: $H_0: \mu = 0$ vs $H_1: \mu < 0$ (5% in left tail)



6. Portfolio analysis

What is a value-weighted portfolio?



Each firm's weight = its market cap ÷ total market cap of all firms

Total market cap = \$1000B

Portfolio return $R_p = \sum w_i \times R_i = 50\% \times R_A + 30\% \times R_B + 15\% \times R_C + 5\% \times R_D$

6. Portfolio analysis

Mean return, standard deviation & Sharpe ratio

Mean (μ)

Average return
over the period

$$\mu = \frac{1}{T} \sum_{t=1}^T R_t$$

*"How much do we earn
on average?"*

Std Dev (σ)

Spread of returns
around the mean

$$\sigma = \sqrt{\frac{1}{T-1} \sum (R_t - \mu)^2}$$

*"How bumpy is
the ride?"*

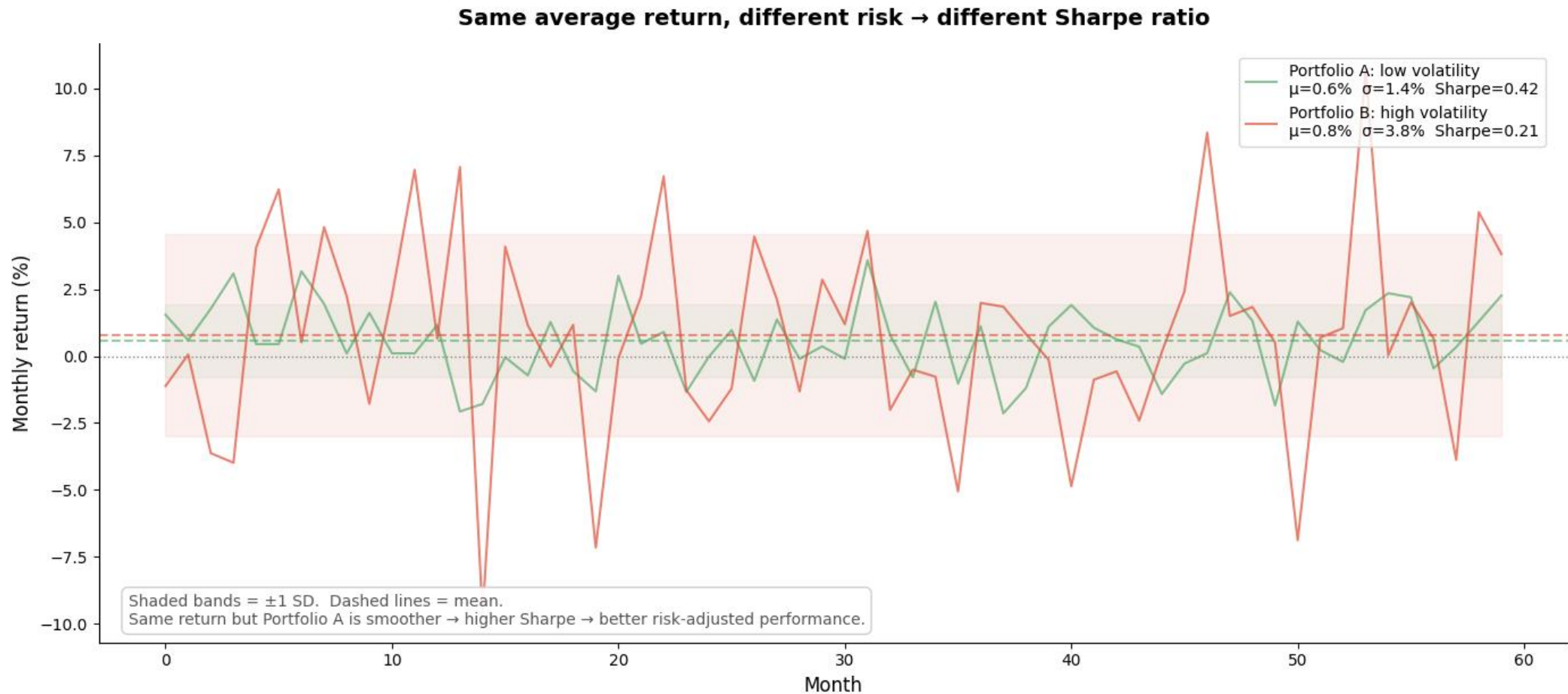
Sharpe Ratio

Return per unit
of risk

$$S = \frac{\mu - R_f}{\sigma}$$

*"Is the return worth
the risk?"*

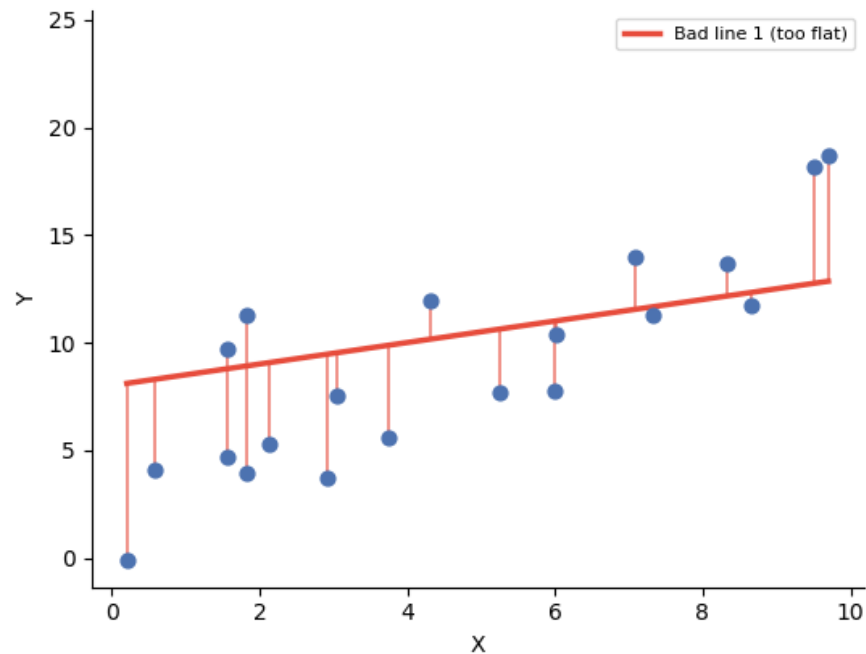
6. Portfolio analysis



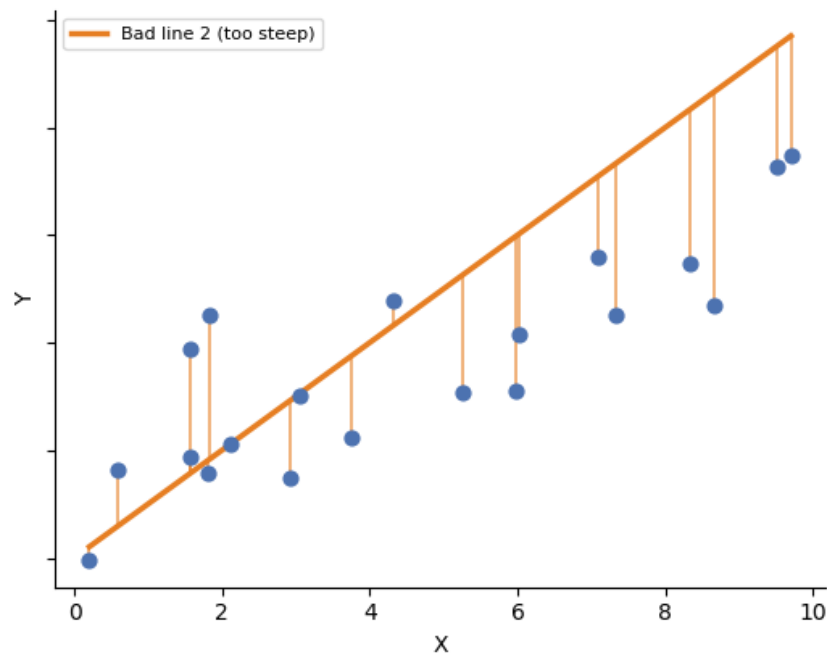
7. Regression - univariate

OLS: finding the line that minimises the Sum of Squared Residuals

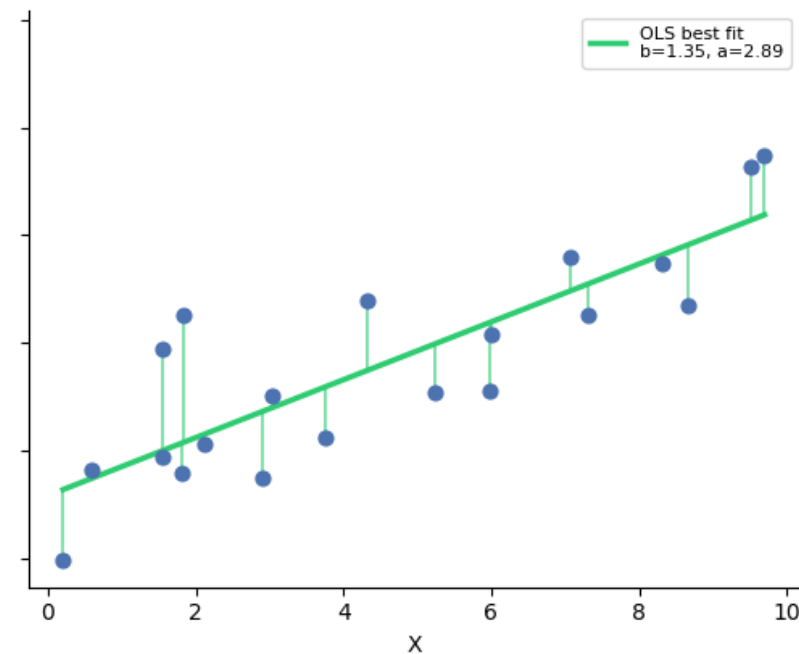
Bad line 1 (too flat)
SSR = 298.8



Bad line 2 (too steep)
SSR = 494.3

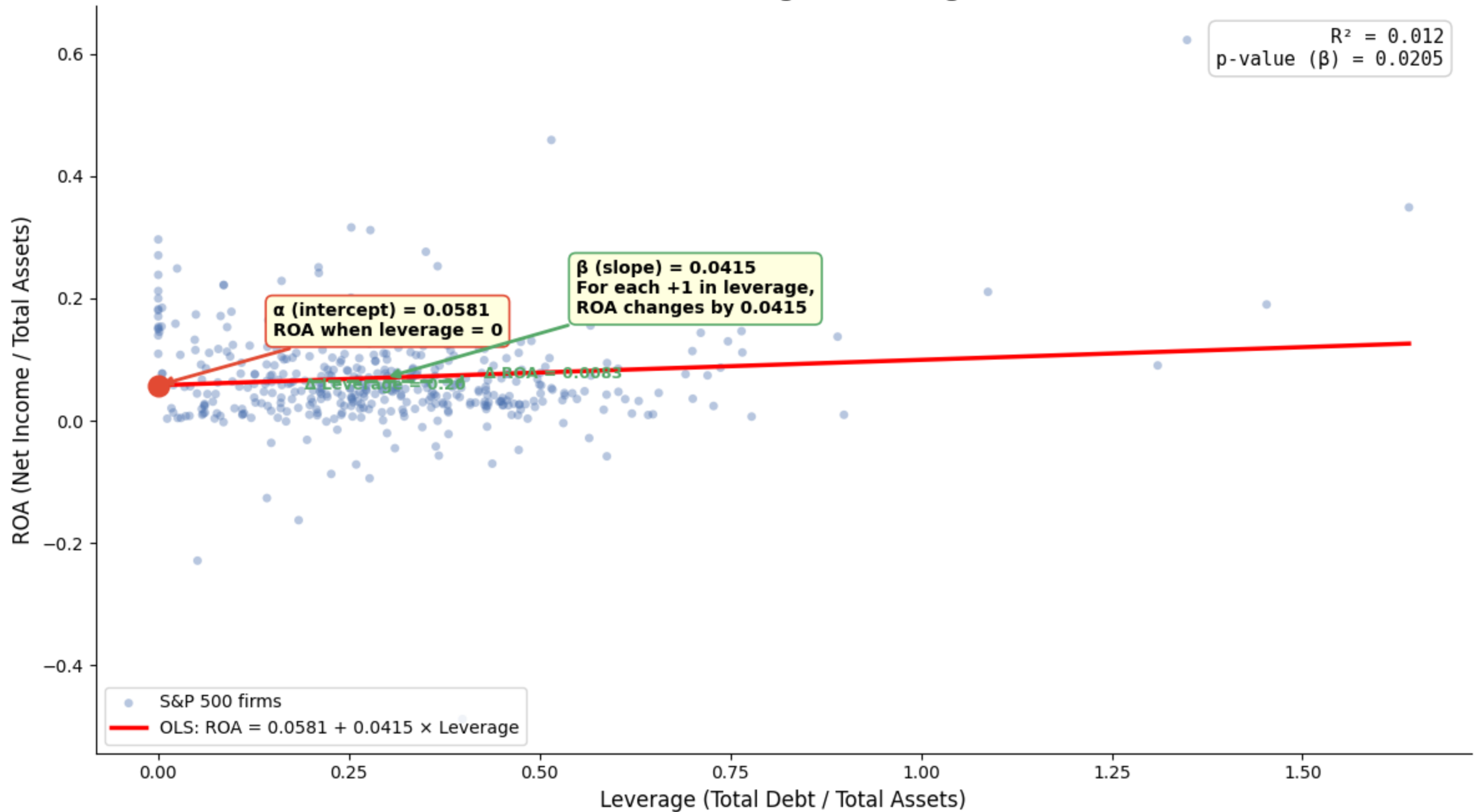


OLS line — minimum SSR
SSR = 139.6



$$SSR = \sum_{i=1}^n e_i^2$$

S&P 500: ROA vs Leverage — OLS regression line



7. Regression - multivariate

$$ROA_i = \alpha + \beta_1 \cdot Leverage_i + \beta_2 \cdot ESG_i + \varepsilon_i$$

The coefficient on *Leverage* in the regression *ROA* on *Leverage* + *ESG* is the same as:

- The coefficient on *Leverage* in the regression of *ROA* on *Residual Leverage*.
- Whereby *Residual Leverage* is the residual of the regression of *Leverage* on *ESG* (it captures the variation of *Leverage* not associated with variation in *ESG*).

7. Regression - multivariate

```
=====
MODEL A: ROA ~ Leverage + ESG (multiple regression)
=====

                        OLS Regression Results
=====
Dep. Variable:          roa      R-squared:                0.015
Model:                  OLS      Adj. R-squared:            0.011
Method:                 Least Squares      F-statistic:          3.334
Date:                  Fri, 27 Feb 2026     Prob (F-statistic):    0.0366
Time:                  11:40:20      Log-Likelihood:       482.66
No. Observations:      428      AIC:                  -959.3
Df Residuals:          425      BIC:                  -947.1
Df Model:              2
Covariance Type:       nonrobust
=====

               coef      std err          t      P>|t|      [0.025      0.975]
-----
const         0.0835      0.020      4.213      0.000      0.045      0.123
leverage      0.0414      0.018      2.291      0.022      0.006      0.077
esg          -0.0089      0.007     -1.315      0.189     -0.022      0.004
=====

Omnibus:            103.748      Durbin-Watson:           2.055
Prob(Omnibus):      0.000      Jarque-Bera (JB):       2274.975
Skew:               0.396      Prob(JB):               0.00
Kurtosis:           14.267      Cond. No.               17.5
=====
```

Coef on *Leverage* is 0.04
(p-value = 0.022)

7. Regression - multivariate

```
=====
MODEL B (Step 1): Leverage ~ ESG (auxiliary regression)
=====

                        OLS Regression Results
=====
Dep. Variable:          leverage    R-squared:                0.003
Model:                  OLS         Adj. R-squared:            0.001
Method:                 Least Squares   F-statistic:              1.262
Date:                   Fri, 27 Feb 2026   Prob (F-statistic):       0.262
Time:                   11:40:20         Log-Likelihood:           59.696
No. Observations:       428             AIC:                     -115.4
Df Residuals:           426             BIC:                     -107.3
Df Model:               1
Covariance Type:        nonrobust
=====

               coef    std err          t      P>|t|      [0.025    0.975]
-----
const         0.2457     0.052     4.737     0.000     0.144     0.348
esg            0.0203     0.018     1.123     0.262    -0.015     0.056
=====

Omnibus:            196.476   Durbin-Watson:           2.037
Prob(Omnibus):      0.000   Jarque-Bera (JB):        1304.103
Skew:               1.856   Prob(JB):                6.57e-284
Kurtosis:           10.704   Cond. No.                16.3
=====
```

Residual Leverage = Leverage – (0.2457 + 0.0203 × ESG)

```
=====
MODEL B (Step 2): ROA ~ Residual Leverage
=====

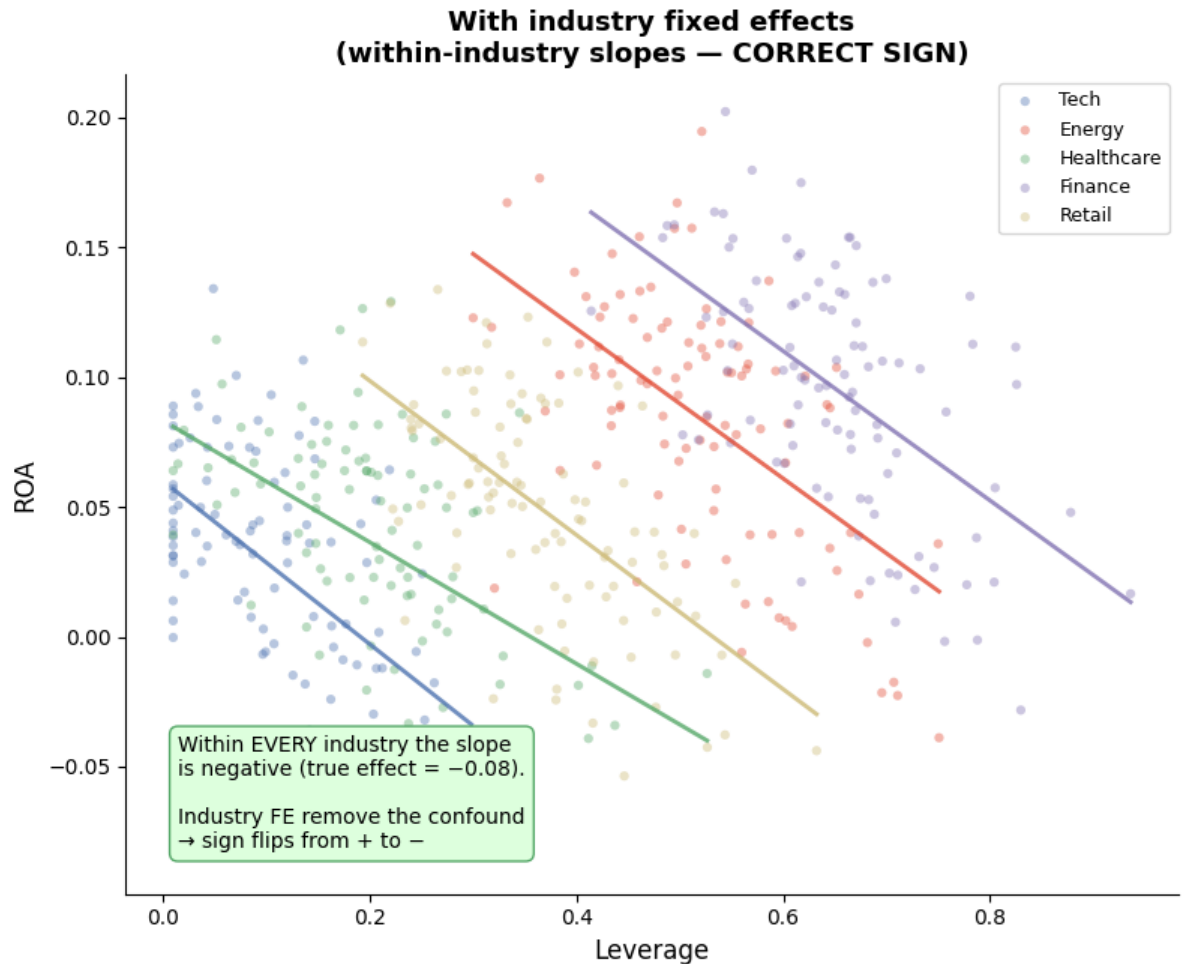
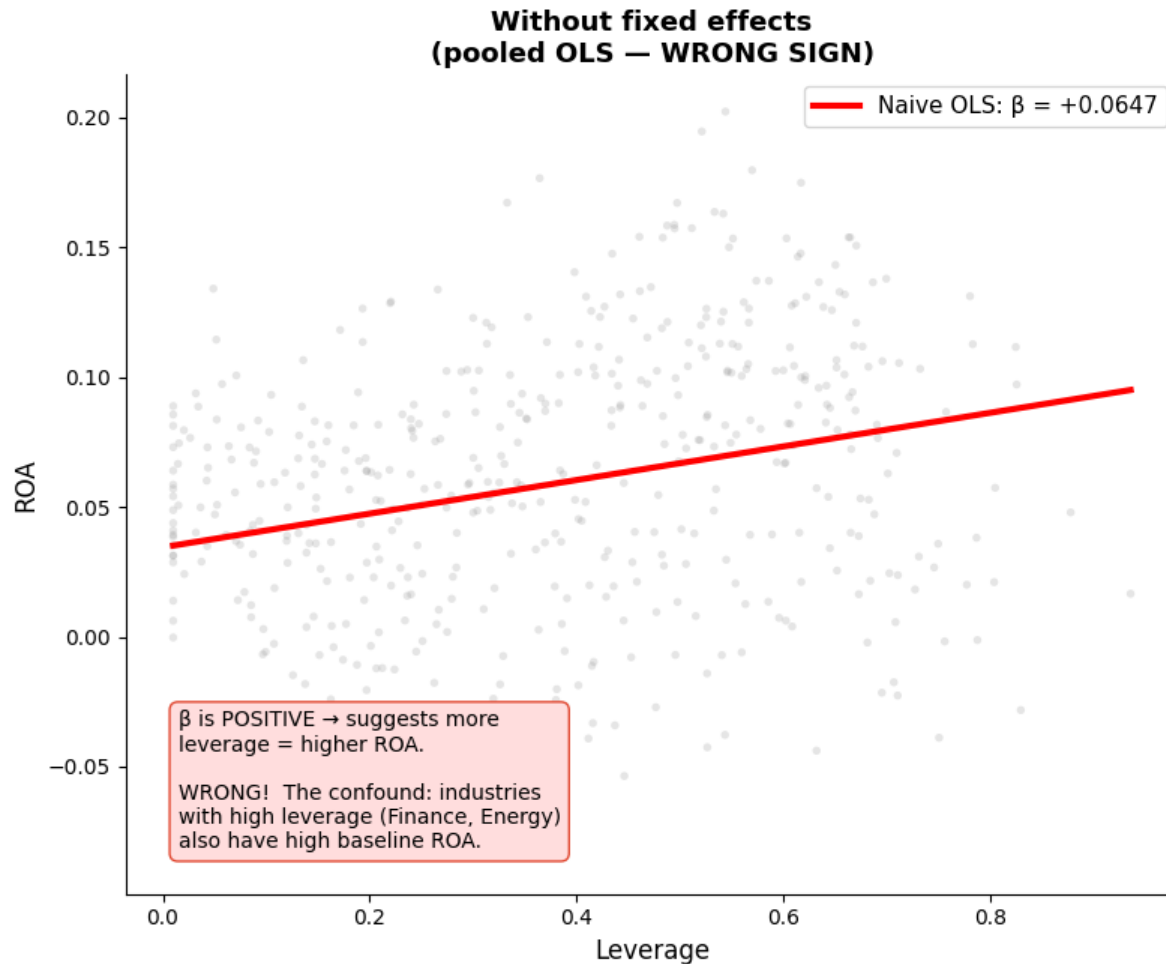
                        OLS Regression Results
=====
Dep. Variable:          roa         R-squared:                0.012
Model:                  OLS         Adj. R-squared:            0.010
Method:                 Least Squares   F-statistic:              5.242
Date:                   Fri, 27 Feb 2026   Prob (F-statistic):       0.0225
Time:                   11:40:20         Log-Likelihood:           481.95
No. Observations:       428             AIC:                     -959.9
Df Residuals:           426             BIC:                     -951.8
Df Model:               1
Covariance Type:        nonrobust
=====

               coef    std err          t      P>|t|      [0.025    0.975]
-----
const         0.0711     0.004    18.701     0.000     0.064     0.079
resid_leverage 0.0414     0.018     2.290     0.023     0.006     0.077
=====

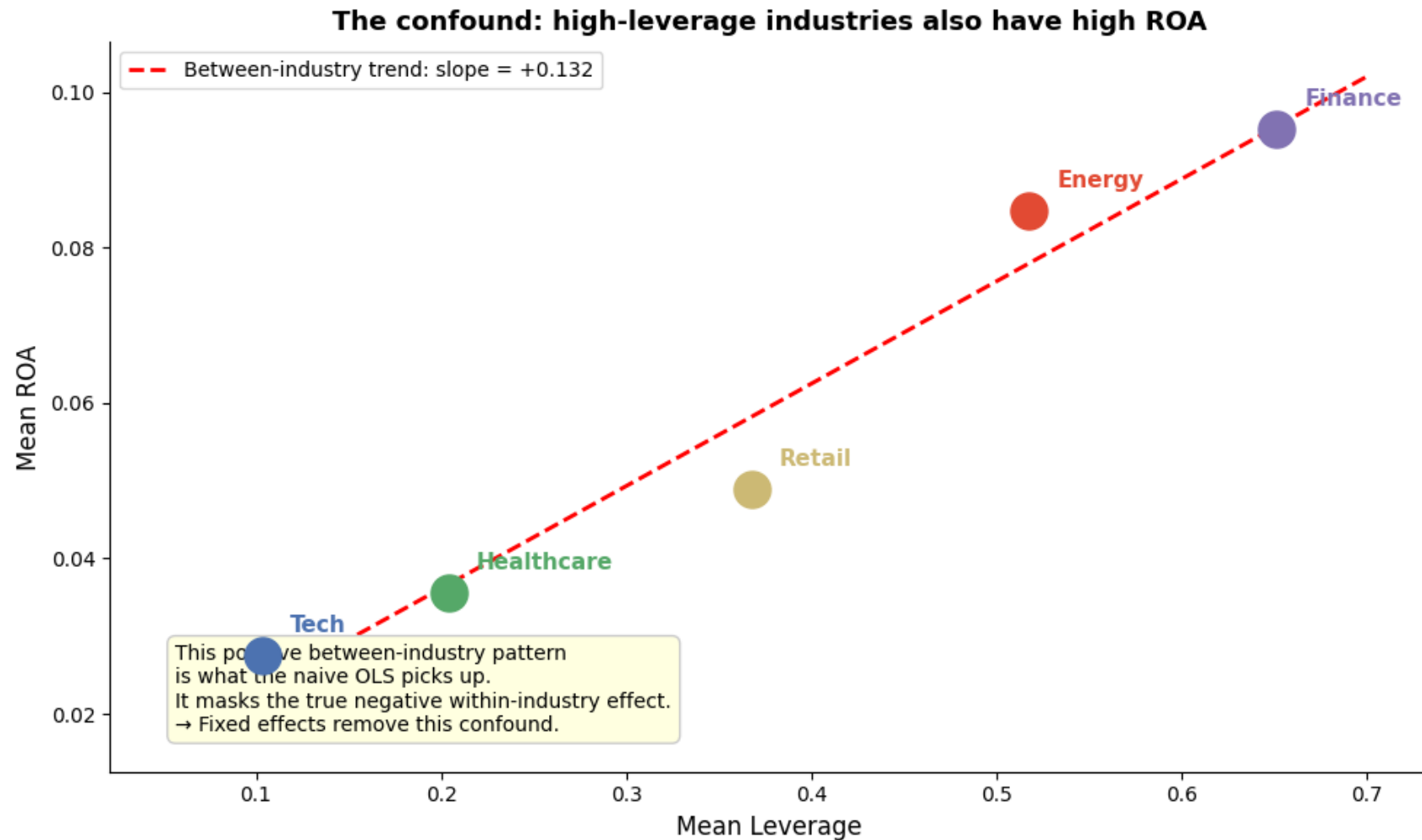
Omnibus:            106.686   Durbin-Watson:           2.057
Prob(Omnibus):      0.000   Jarque-Bera (JB):        2311.905
Skew:               0.444   Prob(JB):                0.00
Kurtosis:           14.351   Cond. No.                4.75
=====
```

Coef on Residual Leverage is 0.04 (p-value = 0.022)

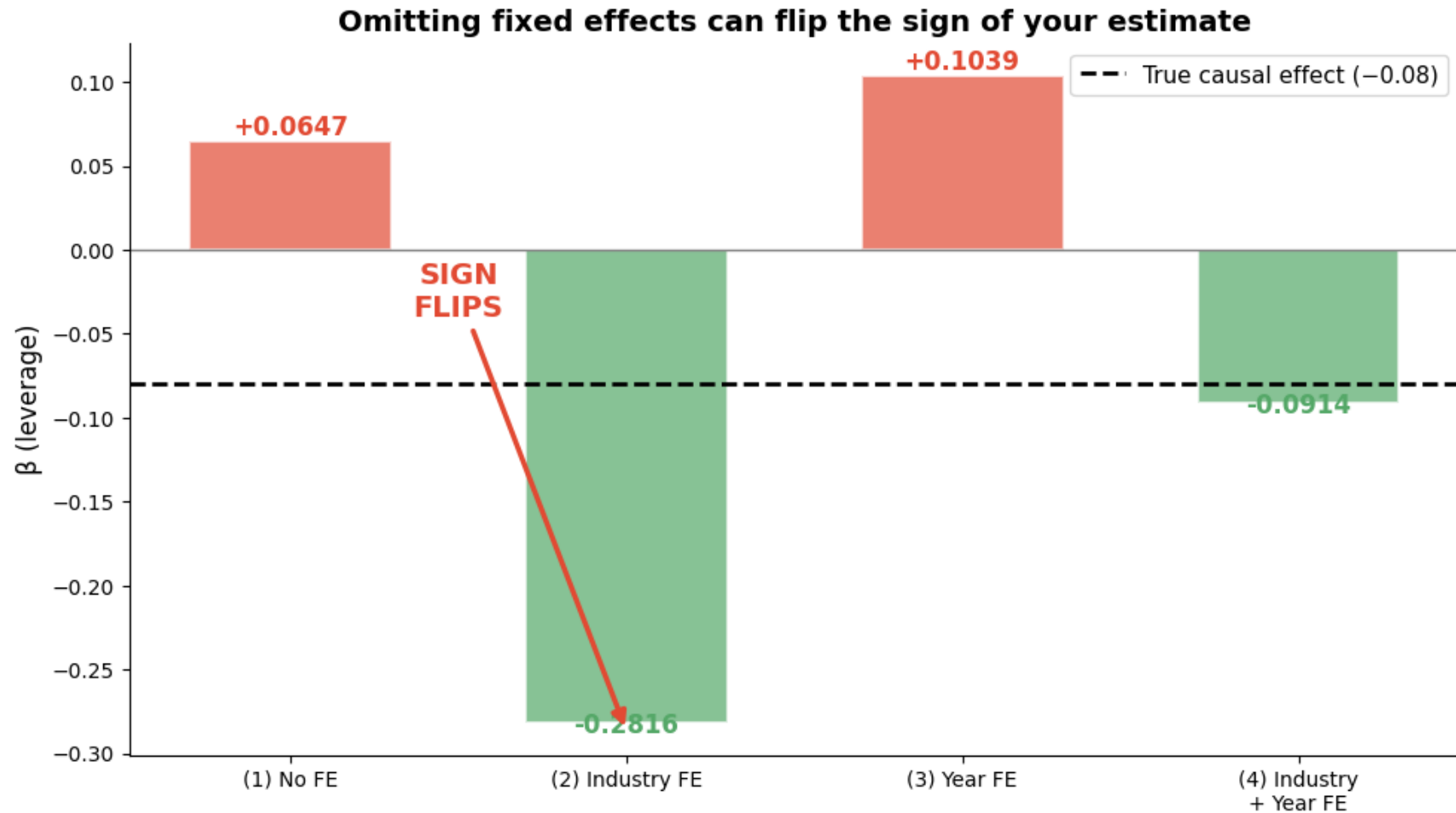
7. Regression – fixed effects



7. Regression – fixed effects



7. Regression – fixed effects



Red = wrong sign (positive) | Green = correct sign (negative) | Dashed = true effect (−0.08)

8. Event study

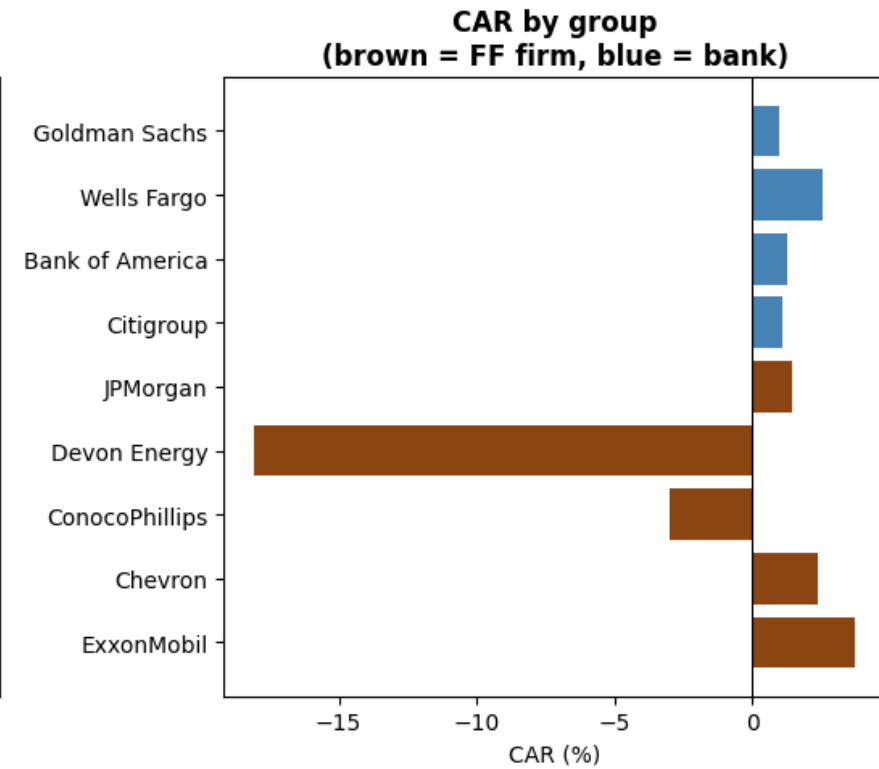
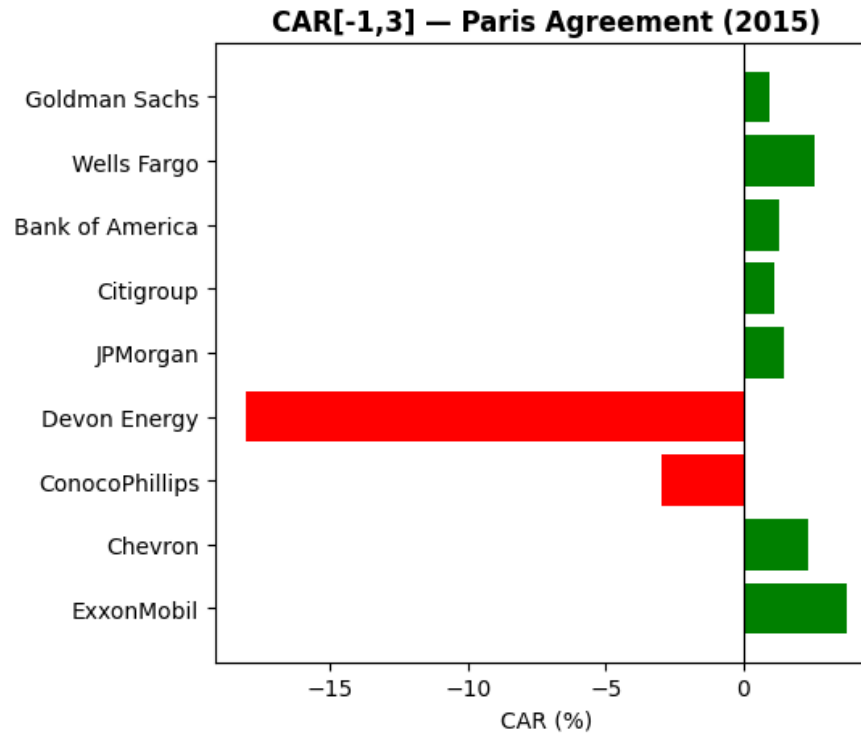
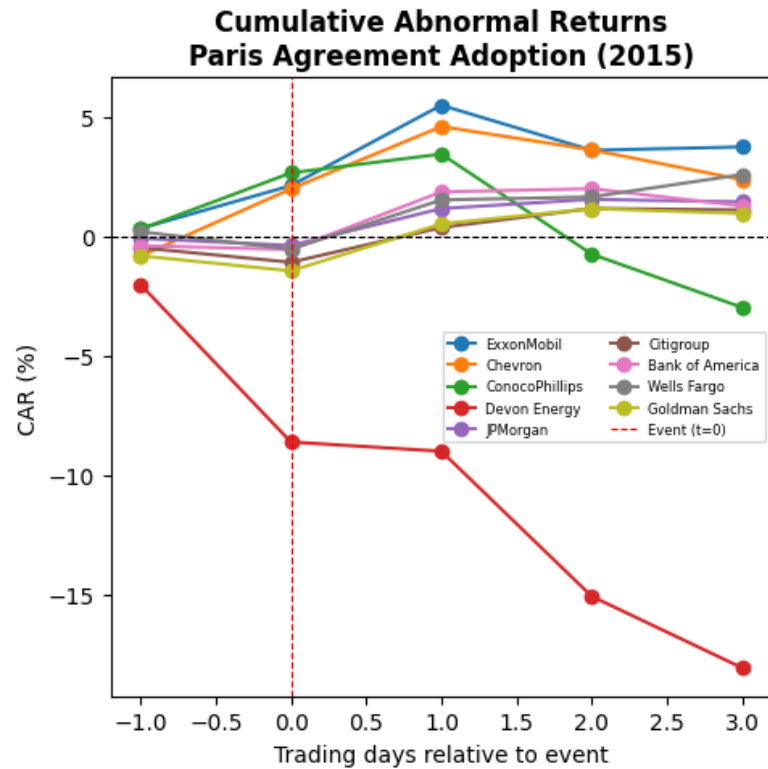
Abnormal reaction of a stock to an event:

STEP 1. Collect data

STEP 2. Compute abnormal returns (in excess of market model)

STEP 3. Cumulate abnormal returns over an event window

8. Event study



9. Cursor

See in class demo